Modulation

Analogue and Digital Communications

Lecture 1

19 August 2024

Textbook:

Modern Digital and analogue communication system
3rd edition by BP Lathi

Block diagram of communication system:
(Simpler communication system):

Center of block diagram

Input message → Input transducer → Input signal → Transmitter → Transmitted signal → Channel → Received signal → Receiver → Output signal → Output transducer → Output message

Distortion and noise → Channel

Input message ↓ Human Voice, Television, pictured data

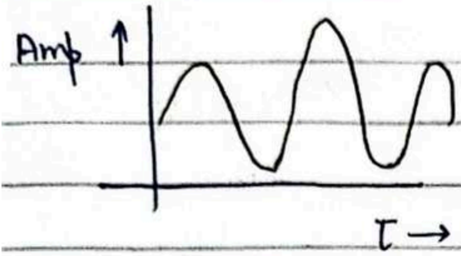
Input signal ↳ Baseband signal (Electrical waveform)

SNR (Signal to noise ratio) is the ratio of signal power to noise power. Noise accumulates along the path. Noise level increases with the distance from transmitter

* No communication signal work with modulation.
* The signal on transmitter side should reach with same accuracy to receiver side
* The thing you are transmitting you need to recover
* **Modulation** process of superimposing a low frequency signal on a high frequency carrier signal ↳ transmit signal efficiently
* **Transducer** convert variation in physical quantities like pressure or brightness into an electrical signal
* **Baseband** convert input into electrical signal
* **Channel** work as medium transmit signal it could be wireless or copper wire

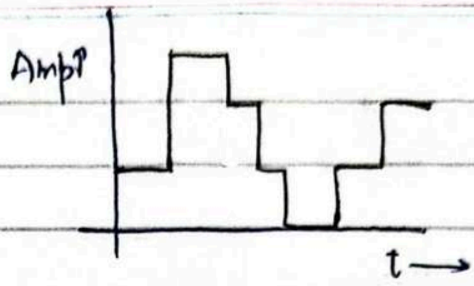
Distortion and noise (this is reason we need communication system to produce better form signal

* Larger the signal Larger the noise is
* Channel is a complicated (all the distance signal have to be transfer)
* If we have modulator we need demodulator



Analogue Signal

Message define on a continuum (eg microphone)



Digital Signal

Finite set of possible messages (e.g Typewriter)

Analogue to Digital Conversion

Analogue to digital conversion of signal Involves the steps

1. Sampling (time axis)
2. Quantization (amplitude axis)
3. Code Assignment to quantized level (we can assign codes at specific level)

Continuum something that keep on going changing slowly over time

Difference b/w both signal is after input signal we can convert analogue to digital if we converted signal before transmitter than it is transmitted signal.

Sampling converting a continuous time signal to a discrete time signal.

After doing these steps we will be able to form PCM scheme

(PCM Scheme - Pulse code Modulation)

Difference b/w analogue & digital communication:

The key feature of a digital communication system is that it deals with finite set of discrete messages. In contrast, to an analogue communication system in which messages are define on a continuum.

Quantization the process of conversion of infinite continuous value into a set of discrete values

Mathematically correct

Lecture 2

21 August, 2024

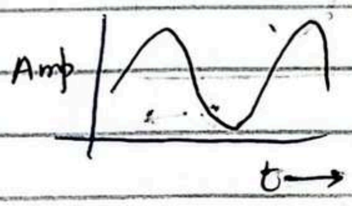
1. Analogue Signal

An Analogue signal is any continous time and continous amplitude signal

No restriction of time $S(t)$ $\qquad t \in \mathbb{R}$ (real number)

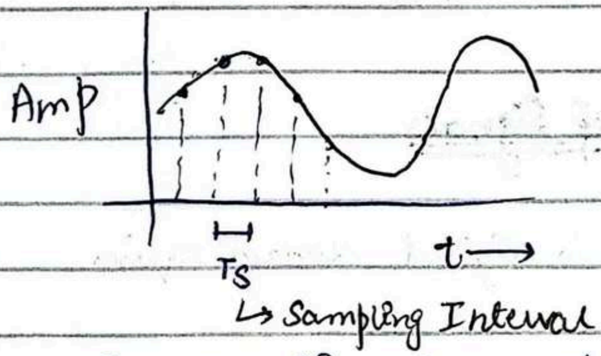
For mathematical treatment

$$S \in \mathbb{C}$$



2. Discrete time signal

These are signals define on a discrete time domain $\{nT_s \mid n \in \mathbb{Z}\}$ and assume continous values on amplitude



Limitation: Time is discrete and continous on Amplitude

$$S(nT_s) \in \mathbb{C}, \; n \in \mathbb{Z}$$

Discrete time signals are derived by sampling at instance that are multiple of T_s.

* Time would be in real domain
* Discrete time signal is it digital signal
* No its not digital signal
* Analogue time & amplitude both par continous ha uso koi bi value ly sakta ha

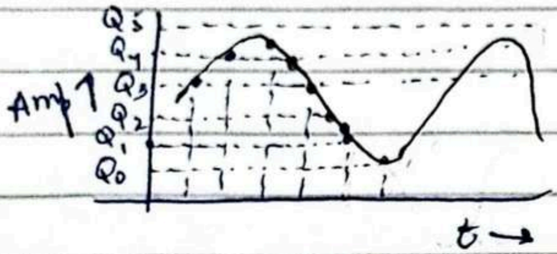
→ Continous values mean ko discretization ka process ha wo time axis par ha

* interval between two signal is sampling interval (T_s)
* $\mathbb{Z}$ is integer
* ~~$n.T_s$ means~~ takes all values +ve & ~~-ve values~~ also
* Restriction is on time axis as it is discrete on it
* $s(t)$ mean time on all time ko bi time dy sakta nT_s sa restrict diya ha

Analogue no restriction of time
discrete restriction of time
but in both amplitude is not restricted

3. Digital Signal

A digital signal is a discrete signal in both time and amplitude domains so a digital signal satisfied.



$$S(nT_s) \in \mathcal{A}, \; n \in \mathbb{Z}$$

$$\mathcal{A} = \{Q_0, Q_1, Q_2, Q_3 \cdots Q_n\}$$

with $\mathcal{A}$ is set of n elements

Discrete time signal na quantization hoti mean amp and time both ain discrete hon ge

we can transmit Q_3 for transmitting that part of signal

Discrete signal time cannot belongs to C because it is now no more continuous

* We are trying to understand mathematical concrete form

* We do sampling first as it is done on time axis and if time axis is discrete than we can make amplitude also discrete which is done by quantization than we rounded off the nearest quantize level.

Analog to digital Conversion of Signals

Conversion of signals from analog to digital domain requires

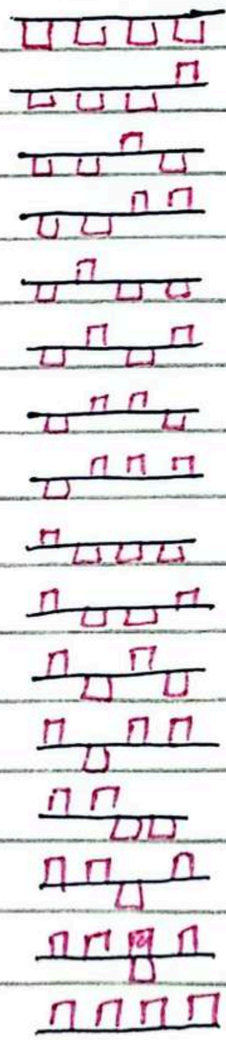
1. Sampling
2. Quantization
3. Code Assignment

The sampling theoram states that if the highest frequency in the signal spectrum is B (refering to bandwidth) (Hertz) (bcz we are taking about frequency f) The signal can be reconstructed from its samples taken at the rate of 2B samples per second

The sample value still lie in a continous range and can take any one of the infinite value in the range.
The problem is resolved by Quantization where every one of the sample is rounded off to the nearest quantized level.

If 16 levels are assigned than we could design a code for the transmission of digits (0 – 15)

Digit	Binary Equivalent	Pulse code waveform
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	
11	1011	
12	1100	
13	1101	
14	1110	
15	1111	



Zero is downward and -ve
one is upward and +ve

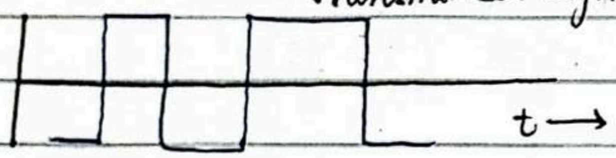
The scheme of transmitting data by digitizing and using pulse codes to transmit the digitize data is known as pulse code modulation. (PCM)

Recived signal is distorted due to noise or may be some large distance is travelled

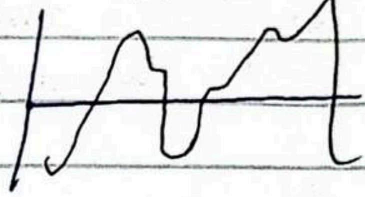
~~Then~~ Major advantage of digital communication is the bossibility of regeneration of signals

Regeneration of signal

Transmitted Signal



Recived signal



Digital signal Ka benefit ya ha ka humay Level pata ha to hum aik level sa dosray level par jatay hn noise ki waja sa level distrot ~~nahi hotay~~

This process is done in ~~digital~~ communication and cannot be done in analog bcz it is continous and if noise is added so it is difficult to differentiate

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Lecture 3

Shannon's Theoram

$$C = B\log_2(1+SNR)$$

$C \rightarrow$ Channel capacity
$B \rightarrow$ Bandwidth
$SNR \rightarrow$ Signal to noise ratio

* Ratio of information transmission (Channel Capacity)

Channel Capacity (C) $\propto$ Bandwidth (B)

* B and SNR are exchangeable

To maintain a given rate and accuracy of transmission we trade SNR for bandwidth but not vice versa.
but we donot trade bandwidth for SNR

$$SNR_2 \approx SNR_1 \frac{B_1}{B_2}$$

So we say
$$SNR_2 \simeq SNR_1 \frac{B_1}{B_2}$$

Let take an example

$SNR_1 = 64 \quad B_1 = 4 \Rightarrow B_2 = 8$

$SNR_2 = SNR_1 \frac{B_1}{B_2}$

$= 64 \frac{4}{8}$

$\boxed{SNR_2 = 8}$

* Bandwidth
physical parameter
which define frequency
* Kitna band na jo transmit karna ha

* SNR is a physic paramet ratio signal to noise
in the same unit having
same ratio
$\frac{\text{signal power}}{\text{noise power}}$

* If we try to built a relationship b/w capacity and bandwidth it is directly proportional
* SNR and B are not direct relationship not even inverse relationship

* rate of information is directly proportional to bandwidth and is exchangable with SNR

* we double the bandwidth

[Small increase in B buys a large advantage in SNR]

* C is not physical parameter it is a mathematical parameter
* B is physical quantity
* SNR is very much a physical quantity

Rate of information transmission or channel capacity C in simplest words gives the maximum number of bits that can be transmitted per second with probability of error approaches to zero

$P_e \longrightarrow 0$

* 100 bits ⟶ 1 bit error

$P_e = \frac{1}{100}$

If $N = 0$ then $C \rightarrow \infty$

If Noise = 0
if N = 0 so probability of error approaches to 0 by which C approaches to ∞

Shannon's theorem represents the upper limit on rate of communication over a channel and cannot be practically achieved as the time delay and complexity ⟶ ∞

* Time delay really large complexity exceed in practical

Primary Communication Sources

* Bandwidth critical parameter expensive limited source

There are two primary communication sources which are

→ Band width (B)

→ Transmitted power (SNR in this case)

* SNR not transmitted power it is actually Signal to noise power

* Transmitted power is also limited Tower is to transmit to large distances but this would also be in limit

* communication system is design for balancing Band width & SNR

* Importance of Bandwidth
(In designing use efficiently as possible)

* Frequency division multiplexing

F_1 F_2 F_3 F_4 F_5 F_6 F_7 F_8

100 ⟼ 2 Hz 8 signal

Significance of Channel Capacity (C):

If the information rate $R < C$ than it is theoretically possible to achieve reliable transmission [probability of error approaches to zero $P_e \longrightarrow 0$] through the channel by appropriate channel coding

If the information rate $R > C$ than reliable transmission is not possible there is no way of achieving reliable transmission

* Channel coding
↓
Parity Bit
Use of parity (to protect data)
Even parity data is evenly distributed and if not even data than there is error

* 32 bit of information & 2 bit of parity this ensure correctness of data & protection of data of 30 bits

If $R > C$
* P_e suddenly increase there is huge error

We want to reduce error (fundamental purpose)
* If channel capacity limit exceed than its fundamental purpose destroyed

Signal Energy and Energy spectral Density

$$E_g = \int_{-\infty}^{\infty} g^2(t)\, dt$$

Spectral & spectrum
↳ frequency

For complex signal $E_g = \int_{-\infty}^{\infty} |g(t)^*|^2 dt$

In communication we are interested in frequency bcz it define bandwidth

The energy E_g of a signal $g(t)$ is define as the area under $|g(t)|^2$
The signal energy is also determinable for the Fourier transform $G(\omega)$ through the Parseval's theorem.

$$g(t) \Leftrightarrow G(\omega)$$

$$G(\omega) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt$$

$$g(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} G(\omega)\, e^{j\omega t} d\omega$$

$g^*(t)$
↳ conjugate

$$E_g = \int_{-\infty}^{\infty} g(t)\, g^*(t)\, dt$$

$$= \int_{-\infty}^{\infty} g(t)\left[\frac{1}{2\pi}\int_{-\infty}^{\infty} G^*(\omega) e^{-j\omega t} d\omega\right] dt$$

here we open it using Inverse fourier transform

$g^*(t)$ being the conjugate of $g(t)$ can be expressed by modifying the IFT Interchanging the order of integration yields

$$E_g = \frac{1}{2\pi}\int_{-\infty}^{\infty} G^*(\omega)\left[\int_{-\infty}^{\infty} g(t)\, e^{-j\omega t} dt\right] d\omega$$

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega)\, G(\omega)\, d\omega$$

$$\boxed{E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2\, d\omega}$$

this is parseval's Theoram

Energy Spectral Density (ESD) $\Psi_g(\omega)$

* connected to signal energy as $|G(\omega)|^2$

$$\Psi_g(\omega) = |G(\omega)|^2$$

28 August 2024

Lecture

Signal Energy and Energy Spectral Density (ESD)

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 \, dt$$

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 \, d\omega$$

Energy spectral Density (The energy per unit bandwidth) (ESD)

$$\Psi_g(\omega) = |G(\omega)|^2$$

Example Find the signal energy for

$$g(t) = e^{-at} u(t); \; a > 0$$

by time domain specification and the equivalent signal Spectrum $G(\omega)$

Solution:

$$g(t) = e^{-at} u(t)$$

$$G(\omega) = \frac{1}{j\omega + a}$$

$u(t) \rightarrow$ changes lim from 0 to ∞

$$E_g = \int_{-\infty}^{\infty} g^2(t) \, dt$$

$$= \int_{0}^{\infty} e^{-2at} \, dt$$

$$\boxed{E_g = \frac{1}{2a}} \rightarrow (A)$$

if parseval theorem satisfy both time domain and energy domain will get same

E_g by the Parseval's theorem

$$E_g = \frac{1}{2\pi}\int_{-\infty}^{\infty} |G(\omega)|^2 \, d\omega$$

$$E_g = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{\omega^2 + a^2} \, d\omega$$

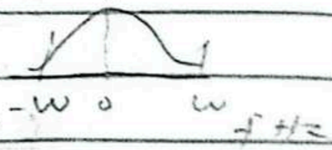
$$= \frac{1}{2\pi a} \tan^{-1} \frac{\omega}{a} \Big|_{-\infty}^{\infty}$$

$$\boxed{E_g = \frac{1}{2a}} \quad \text{—— (B)}$$

Hence the Parseval's theorem is verified and the signal energy E_g can be founded by both time domain $g(t)$ and its fourier transform $G(\omega)$.

Essential Bandwidth W of the signal

For most practical signals the signal spectrum beyond W Hz can be supressed with little effect on the signal shape and energy.
The bandwidth W is called the essential bandwidth of the signal



limit the spectrum upto W Hz that are to recover signal shape & energy

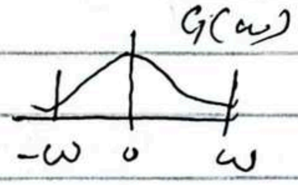
Imp

Example Estimate the essential bandwidth W rad/s of the signal $e^{-at}u(t)$ if the essential band is required to contain 95% of signal energy

$$e^{-at}u(t) \iff \frac{1}{j\omega + a}$$

and the ESD $\Psi_g(\omega) = |G(\omega)|^2 = \frac{1}{\omega^2 + a^2}$

$$E_g = \frac{1}{2a}$$



$$E_g = \frac{1}{2\pi} \times \text{area under ESD}$$

$$\frac{1}{2a} = \frac{1}{2\pi} \times \text{area under ESD}$$

Essential band width → frequency domain

$$0.95 \times \frac{1}{2a} = \frac{1}{2\pi}\int_{-W}^{W} \frac{1}{\omega^2 + a^2}\, d\omega$$

Spectral ka dekh sa dakh to kitna spectrum important ho

$$\frac{0.95}{2a} = \frac{1}{2\pi a} \tan^{-1}\frac{\omega}{a}\Big|_{-W}^{W}$$

$$\frac{0.95}{2a} = \frac{1}{\pi a}\tan^{-1}\left(\frac{W}{a}\right)$$

$$\frac{0.95\pi}{2} = \tan^{-1}\left(\frac{W}{a}\right)$$

$W = 12.706a$ rad/s

Spectral components of $g(t)$ from 0 to $12.706a$ rad/s ($2.02a$ Hz) contribute 95% of signal energy

Signal Power & power Spectral Density (PSD)

signal power when energy become infinite & cannot calculate "signal energy" So power energy come. Eg periodic signal

For power signals [Signals with $g(t) \nrightarrow 0$ as $t \to \infty$] the signal power is meaningful as "time average of energy" averaged over an infinite time interval

The power P_g of a signal $g(t)$ is given by

$$P_g = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

So the signal power is thus just the time average of signal energy

Power Spectral Density (PSD) $S_g(\omega)$

$$S_g(\omega) = \lim_{T \to \infty} \left| \frac{G_T(\omega)}{T} \right|^2$$

and consequently

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega = \frac{1}{\pi} \int_{0}^{\infty} S_g(\omega) d\omega$$

2 Sep 2024

Lecture 5

Signal Power and Power Spectral Density (PSD)

$$P_g = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

$$S_g(\omega) = \lim_{T \to \infty} \frac{|G_T(\omega)|^2}{T}$$ power spectral density)

and consequently

for even function: $\boxed{P_g = \frac{1}{\pi} \int_0^{\omega} S_g(\omega) d\omega}$

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega$$ (function other than even function

The power is $\frac{1}{2\pi}$ times the area under PSD

PSD is a positive real and even function of ω

The PSD $S_g(\omega)$ represents the power per unit bandwidth BW (in Hertz) of the spectral components at frequency ω

Physical interpretation of power spectral Density

The power contributed by the spectral components within the band ω_1 to ω_2 is given by

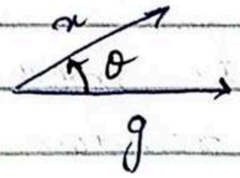
$$\Delta P_g = \frac{1}{\pi} \int_{\omega_1}^{\omega_2} S_g(\omega) d\omega$$

Time auto relation function

CamScanner

Correlation

Two vectors g and x are correlated if g has a large component along x



Simplest way → smaller the θ larger the correlation

assume $\theta = 0^\circ$ x and g would start overlap than there correlation is max

Correlation coefficient

in vector form

$$C_n = \cos\theta = \frac{g \cdot x}{|g|\,|x|}$$

$$-1 \leq C_n \leq 1$$

* Correlation coefficient is trying to quantify correlation

→ For aligned vector $(C_n = 1)$ $\theta = 0$

→ For vector in opposite direction $(C_n = -1)$ $\theta = 180$

→ For vector orthognal $(C_n = 0)$

For signals $g(t)$ and $x(t)$

$$C_n = \frac{1}{\sqrt{E_g E_x}} \int_{-\infty}^{\infty} g(t)\,x(t)\,dt$$

$C_n = 0$ if $g(t)$ and $x(t)$ are orthognal

Autocorrelation:

The correlation of a signal with itself is called autocorrelation

* we take first original signal than shifted signal

Time domain →

$$\psi_g(\tau) = \int_{-\infty}^{\infty} g(t)\,g(t-\tau)\,dt$$

$$\psi_g(\tau) \Leftrightarrow \Psi_g(\omega)$$ (gives ESD) Energy spectral density

Auto correlation method for a signal $g(t)$ the energy spectral density is $|G(\omega)|^2$ can also be found by taking the fourier transform of its autocorrelation function

The reason for this auto alternate method is to lay a foundation for dealing with random signal

Random signal noise signal is difficult to handle.

The time auto correlation function $R_g(t)$ of a real power signal $g(t)$ is defined as

$$R_g(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} g(\tau) g(t+\tau)\, dt$$

$R_g(\tau)$ is an even function of τ this means for real $g(t)$ $R_g(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t)\, g(t-\tau)\, dt$

$$R_g(\tau) = R_g(-\tau)$$

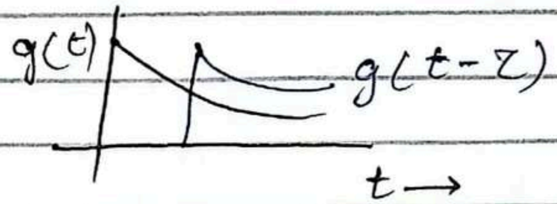
Q. Find the time autocorrelation function of signal $g(t) = e^{-at}u(t)$ and from it determine the ESD of $g(t)$

①
② Find ESD through auto correlation

$$g(t) = e^{-at}u(t)$$

and

$$g(t-\tau) = e^{-a(t-\tau)}u(t-\tau)$$



Autocorrelation function $\psi_g(\tau)$ is given by area under the product $g(t)\,g(t-\tau)$

$$\psi_g(\tau) = \int_{-\infty}^{\infty} g(t)\,g(t-\tau)\,dt$$

change limit from 0 to ∞

$$= \int_{-\infty}^{\infty} e^{-at}u(t)\, e^{-at}e^{a\tau}\,u(t-\tau)\,dt$$

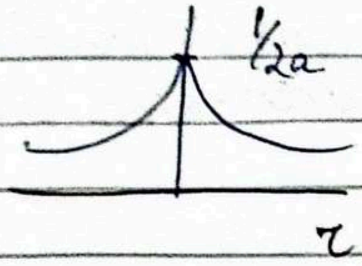
change limit from τ to ∞

$$= e^{a\tau}\int_{\tau}^{\infty} e^{-2at}\,dt$$

$$= \frac{1}{2a}e^{-a\tau} \quad \text{(time autocorrelation)}$$

For real $g(t)$, $\psi_g(t)$ is an even function

$$\psi_g(\tau) = \frac{1}{\;}e^{-a|\tau|}$$



$\psi_g(t)$ (auto correlation)

ESD $\Psi_g(\omega) = \dfrac{1}{\omega^2 + a^2}$ (Fourier transform of $\psi_g(t)$)

$$P_g = \lim \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 \, dt$$

$$S_g(\omega) = \lim_{T\to\infty} \frac{|G_T(\omega)|^2}{T} \quad \text{.... PSD}$$

$$P_g = \frac{1}{\pi} \int_{-\infty}^{\infty} S_g(\omega)\, d\omega \quad \text{(even function)}$$

$$\Delta P_g = \frac{1}{\pi} \int_{\omega_1}^{\omega_2} S_g(\omega)\, d\omega \quad \text{(Bandwidth)}$$

Correlation:

$C_n = \cos\theta = \dfrac{g \cdot x}{|g||x|}$

$C_n = 1$ (aligned)

$C_n = -1$ (opposite)

$C_n = 0$ (orthognal)

$$C_n = \frac{1}{\sqrt{E_g E_x}} \int_{-\infty}^{\infty} g(t)\, x(t)\, dt$$

$-1 \leq C_n \leq 1$

Autocorrelation: $\psi_g(\tau) = \int_{-\infty}^{\infty} g(t)\, g(t-\tau)$

4 September

Lecture 6

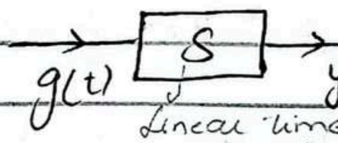
Spectrum
↳ frequency domain

ESD
↳ frequency domain

Energy Spectral Density [ESD] of The input and output of LTI System

If $g(t)$ and $y(t)$ are the input and the corresponding output of a linear invariant system (LTI) then

output → $y(t) = g(t) * h(t)$



$h(t)$
↳ impulse function

Fourier transform of output → $Y(\omega) = G(\omega)\, H(\omega)$

$Y(\omega) = H(\omega)\, G(\omega)$

↳ Fourier transform of impulse function

$$|Y(\omega)|^2 = |H(\omega)|^2 |G(\omega)|^2$$

$g(t) \Longleftrightarrow G(\omega)$

$|G(\omega)|^2 = \Psi_g(\omega)$

This whole concept is applied to linear time invariant system

This shows that

$$\Psi_y(\omega) = |H(\omega)^2|^2 \Psi_g(\omega)$$

ESD ya to input ki hogi ya output ki ya impulse ki nahi hogi

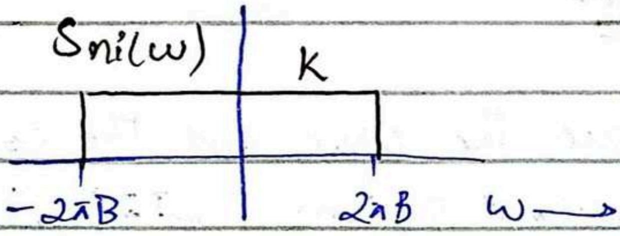
Thus output ESD is $|H(\omega)|^2$ times the input ESD

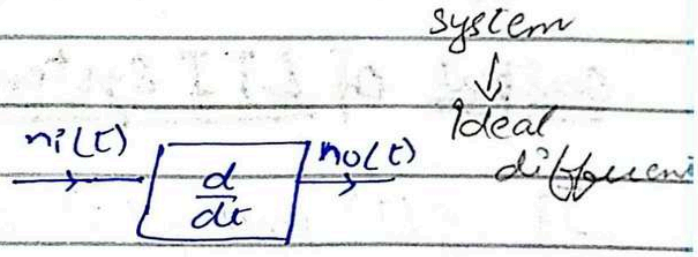
Since the PSD is a time average of ESD, the input/output signal PSD of an LTI system is

$$S_y(\omega) = |H(\omega)|^2 S_g(\omega)$$

A noise signal $n_i(t)$ (input) with PSD $S_{ni}(\omega) = K$ is applied at the input of an ideal differentiator $\frac{d}{dt}$

Determine the PSD and the power of output noise signal $n_o(t)$





The transfer function for an ideal differentiator is

$$H(\omega) = j\omega$$

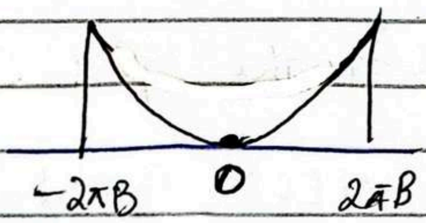
If noise at differentiator output is $n_o(t)$ then

$$S_{n_o}(\omega) = |\underbrace{H(\omega)}_{j\omega}|^2 \underbrace{S_{ni}(\omega)}_{K}$$

$$= |j\omega|^2 K \quad \text{(power spectral density)}$$

↳ spread over a bandwidth

The output PSD is thus parabola



The output noise power N_0 is $\frac{1}{2\pi}$ times area under the output PSD

$$N_0 = \frac{1}{2\pi} \int_{-2\pi B}^{2\pi B} K\omega^2 \, d(\omega)$$

$$= \frac{K}{2\pi} \int_{-2\pi B}^{2\pi B} \omega^2 d\omega$$

$$= \frac{8\pi^2 B^3 K}{3}$$

Lecture 7

9 Sep 2024

Signal Transmission through a Linear System

Fundamental of communicati

Certain things holds for certain time

For a Linear time invariant (LTI) continous time system, the input output relationship is given by

(Time domain cannot do special shaping) $y(t) = g(t) * h(t)$ — (A)

$\rightarrow$ unit impulse response

input — System — Output

Where $g(t)$ is the input, $y(t)$ is the output and $h(t)$ is the unit impulse response

communication system ↳ Band critical parameter ↓ Frequency domain

$$g(t) \Leftrightarrow G(\omega)$$

$$y(t) \Leftrightarrow Y(\omega)$$

$$h(t) \Leftrightarrow H(\omega)$$

where $H(\omega)$ is system transfer function, than the application of the time (Continous)* convolution property yields

This transfer function is use to shape the function ↳ it has amplitude & phase

$$Y(\omega) = G(\omega)\,H(\omega) \rightarrow (B)$$

↳ Linear system

$H(\omega)$ Thus represent here the spectral response of the systems

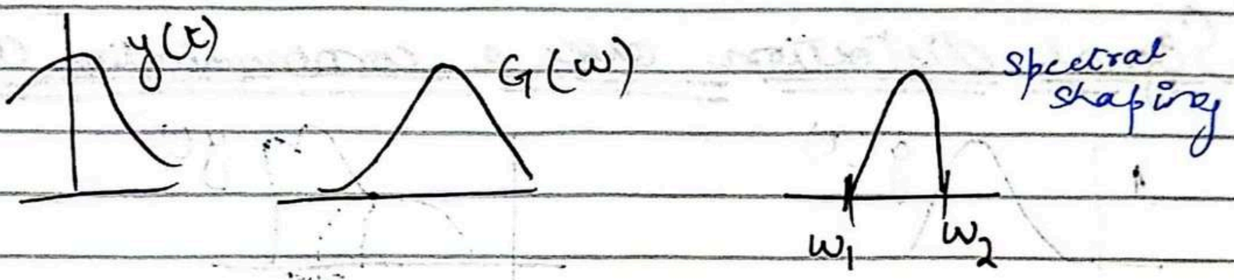
$$|Y(\omega)|\, e^{j\theta_y(\omega)} = |G(\omega)|\,|H(\omega)|\, e^{j[\theta_g(\omega)+\theta_h(\omega)]}$$

Equation Linear system (L.S) brings out the idea of spectral shaping

→ ... bandwidth
→ critical parameter

(or modification) of the signal by the system.

Plots of $|H(\omega)|$ & $\theta_h(\omega)$ as function of ω shows how the system modifies the amplitude and the phase respectively.



Distortionless Transmission:

Transmission is distortionless if input and output have identical waveshape within a multiplicative constant

t_d → time delay

Since $$Y(\omega) = G(\omega)H(\omega)$$

$$H(\omega) = Ke^{-j\omega t_d}$$

$$|H(\omega)| = K \longrightarrow \text{constant}$$

$$\theta_h(\omega) = -\omega t_d \longrightarrow \text{Linear function of } \omega$$

A delayed output that retains the input waveform is also considered distortionless

$$Y(t) = Kg(t - t_d)$$

fixed delay ka sath and fixed const ka sath multiply

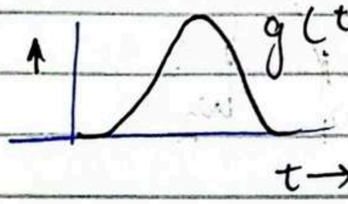
For distortionless case the output signal is the input signal multiplied by K and delayed by t_d

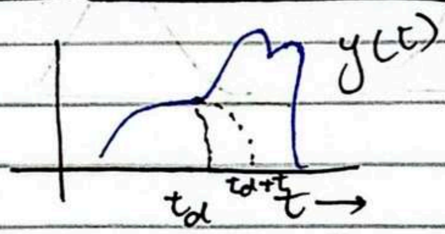
amplitude response 1 hona chahiye, change nahi ho sakta

If delay change
That is not distortionless signal
t_d ki value change nahi ho sakti (delay)

For distortionless transmission t_d should be constant over the band of interest. A system may have flat amplitude response and yet hugely distort a signal if the phase response is not linear (t_d is not const)

Signal distortion over a communication channel





The pulse will spread if the channel characteristics is not ideal that is

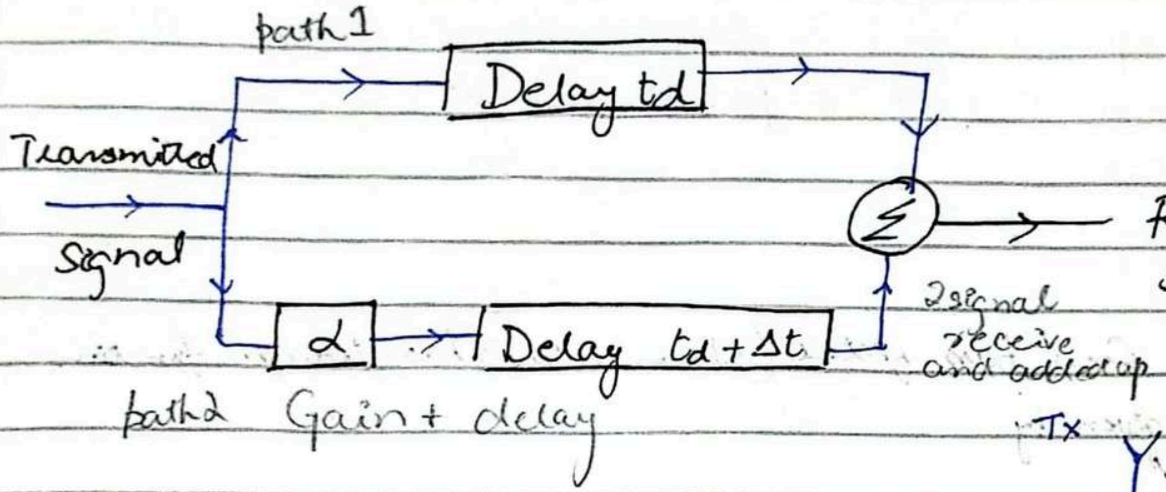
$$\theta_h(\omega) \neq -\omega t_d$$

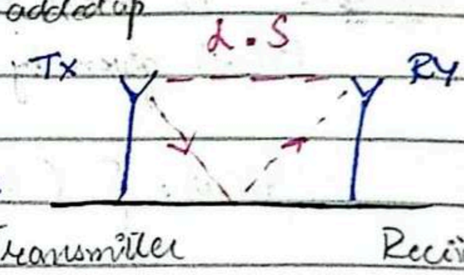
So the spreading or dispersion of the pulse will occur if either the amplitude response or the phase response or both are non ideal

11 September

Lecture 8

Distortion caused by Multipath Effect





Scenario considered has two paths
1) With a unity gain and delay td
2) Other with gain α and delay (td + Δt)

path 1
Line of sight (L.S)
↳ straight line when transmitter & Receiver is visible

path 2
where Reflection occur

Transfer function (fourier transform of unit impulse function)

Transfer function of the two paths are given by $\underbrace{e^{-j\omega t_d}}_{\text{path 1}}$ and $\underbrace{\alpha\, e^{-j\omega(t_d+\Delta t)}}_{\text{gain path 2}}$ respectively

Overall transfer function $H(\omega)$ is

$$H(\omega) = e^{-j\omega t_d} + \alpha \cdot e^{-j\omega(t_d+\Delta t)}$$

$$= e^{-j\omega t_d}(1 + \alpha e^{-j\omega \Delta t})$$

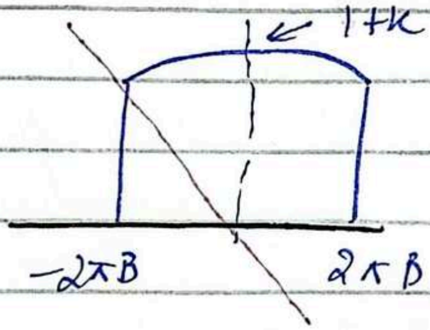
$$= e^{-j\omega t_d}(1 + \alpha \cos\omega_\Delta t - j\alpha \sin\omega\Delta t)$$

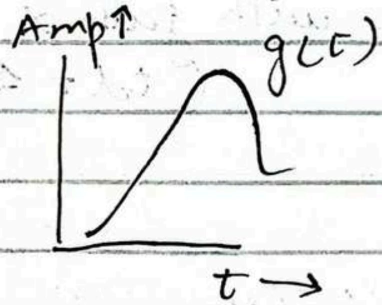
It is not distortionless transmission. It is multipath (where crest & trough cancel each other called signal cancellation)

Both the magnitude and the phase characteristics of $H(\omega)$ are periodic in ω with a period of $\frac{2\pi}{\Delta t}$

Example A low pass filter (shown below) has transfer function $H(\omega)$ given by

$$H(\omega) = \begin{cases} (1 + k\cos T)e^{-j\omega t_d} & |\omega| < 2\pi B \\ 0 & |\omega| > 2\pi B \end{cases}$$





$$\theta_{t_d}(\omega) = -\omega t_d$$

A pulse $g(t)$ bandlimited to B Hz is applied at the input of this filter. Find the output $y(t)$

The filter has ideal phase response and non ideal magnitude response

$g(t) \Leftrightarrow G(w)$; $y(t) \Leftrightarrow Y(w)$

distortion less
↳ both are ideal
if 1 is ideal and other is non ideal or both are non ideal than it has distortion

$$Y(w) = G(w) \cdot H(w)$$
$$= G(w)(1 + K\cos Tw)e^{-jwtd}$$

↳ as in function so remain as it is

$$= G(w)e^{-jwtd} + K[G(w)\cos Tw\, e^{-jwtd}]$$

using time shifting property

$$g(t-t_0) \Leftrightarrow G(w)e^{-jwt_0}$$

$$g(t-T) + g(t+T) \Leftrightarrow 2G(w)\cos Tw$$

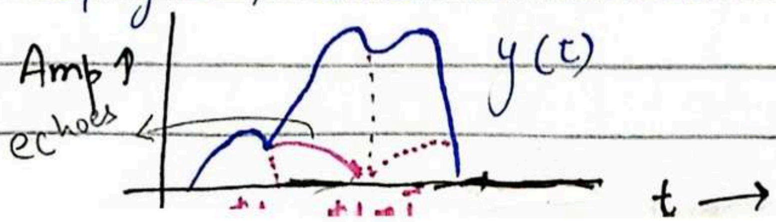
So the time domain equivalent of the equation can be expressed

$$Y(t) = \underbrace{g(t-td)}_{g(t) \text{ delayed by } Td} + \frac{K}{2}\left[\underbrace{g(t-td-T)}_{g(t-T) \text{ delayed by } Td} + g(t-\underbrace{td+T}_{g(t+T) \text{ delayed}})\right]$$

The output $y(t)$ is actually

$$g(t) + \frac{K}{2}\left[g(t-T) + g(t+T)\right] \text{ delayed by } Td$$

It consist of $g(t)$ and its echoes shifted by $\pm\, td$

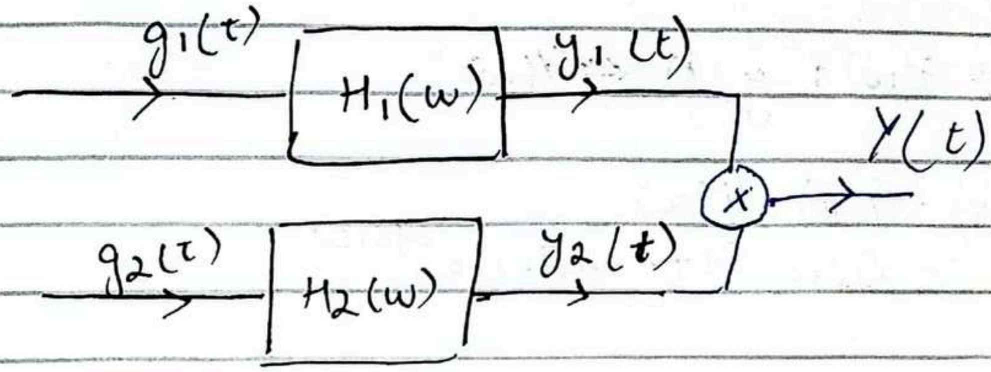


much distorted version of $g(t)$
→ spread out in tir

Lecture 9

16 Sep 2024

Q



Signals $g_1(t) = 10^4 \,\mathrm{rect}(10^4 t)$ and the $g_2(t) = \delta(t)$ are applied at the input of the ideal L.P.F

$$H_1(\omega) = \mathrm{rect}\left(\frac{\omega}{40000\pi}\right) \quad \text{and} \quad H_2(\omega) = \mathrm{rect}\left(\frac{\omega}{2000}\right.$$

The output $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal $Y(t) = Y_1(t)\,Y_2(t)$

$$Y(t) = Y_1(t)\,Y_2(t)$$

a) Sketch $G_1(\omega)$ and $G_2(\omega)$
b) Sketch $H_1(\omega)$ and $H_2(\omega)$
c) Sketch $Y_1(\omega)$ and $Y_2(\omega)$
d) Find the BW of $Y_1(t)$ and $Y_2(t)$ and $Y(t)$

IMP

a) Sketch $G_1(w)$ and $G_2(w)$

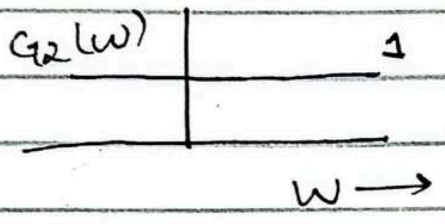
$g_2(t) = \delta(t)$

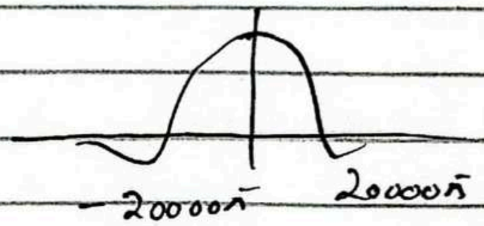
$\Downarrow$

$G_2(w) = 1$

$g_1(t) = 10^4 \operatorname{rect}(10^4 t)$

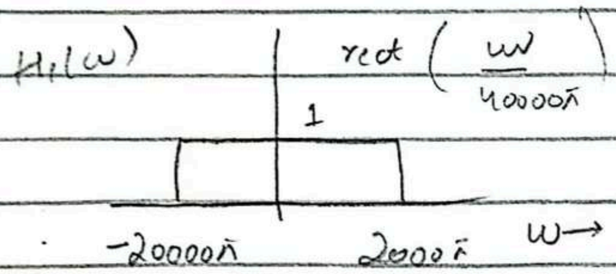
$G_1(w) = \operatorname{sinc}\left(\dfrac{w}{20{,}000}\right)$

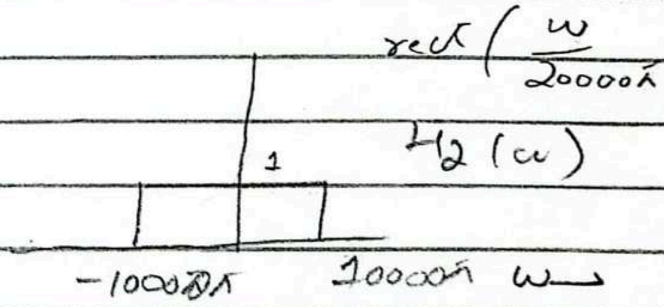




$H_1(w)$ low pass filter it remove other frequency than -2000π & 2000π

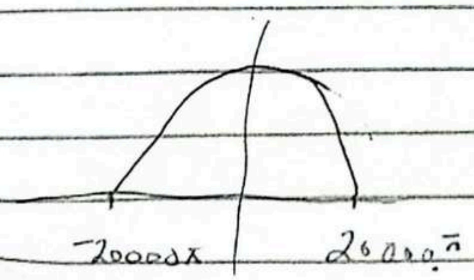
b) Sketch $H_1(w)$ $H_2(w)$

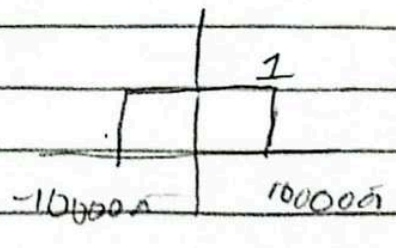




$\rightarrow$ filter will determine its limits

c) Sketch $Y_1(w)$ and $Y_2(w)$





$H_2(w) = G_2(w) H_2(w)$

d)

BW of $Y_1(\omega)$ $= \frac{20000\pi}{2\pi}$ $= 10000 Hz = 10 KHz$

BW of $Y_2 \omega$ $= \frac{10000\pi}{2\pi}$ $= 5000 Hz = 5 KHz$

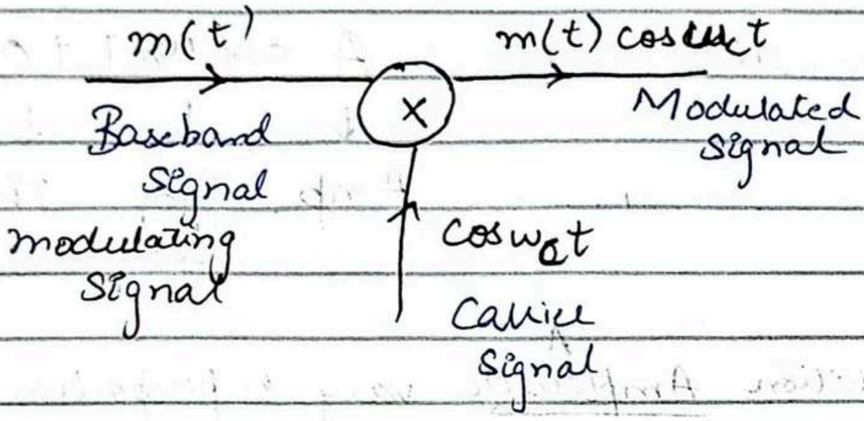
BW of $Y(\omega) = Y_1(\omega) + Y_2(\omega)$
$= 10 + 5$
$= 15 KHz$

Lecture 10 (Chapter 4)

18 September

Amplitude Modulation

→ 2 inputs
Modulating Signal
↓
Low frequency
→ Audio signal
0 — 4 KHz

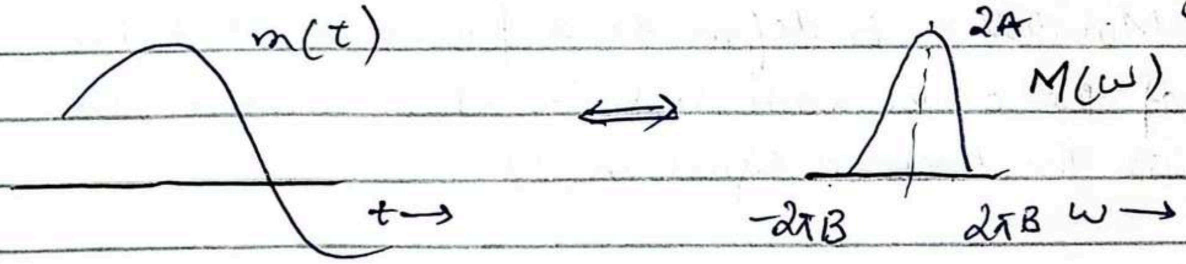


Carrier Signal
↓
must higher frequency

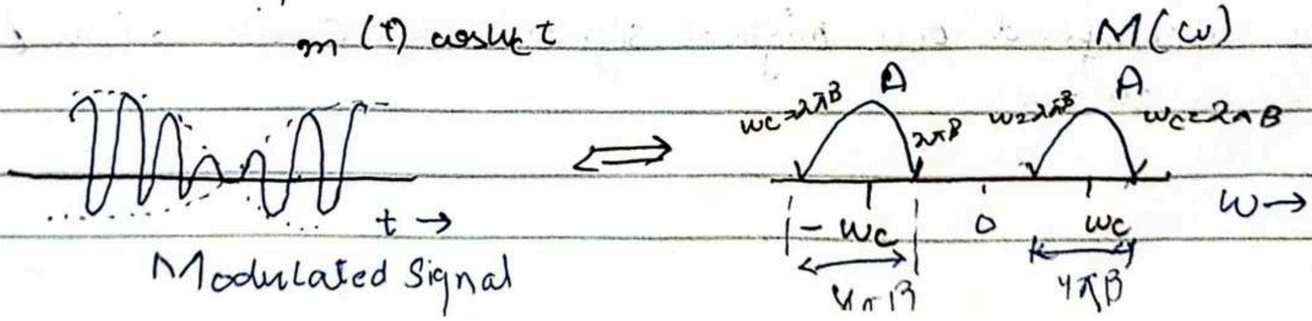
Baseband
↳ any signal generate at input
↳ e.g audio, video

Modulator

Spectrum
↳ Band width
↳ B
↳ ω
↳ 2πB



Modulating Signal
↓
Original signal produce at input



$$m(t) \Longleftrightarrow M(\omega)$$

$$m(t)\cos\omega_c t \Longleftrightarrow \frac{1}{2}\left[M(\omega+\omega_c) + M(\omega-\omega_c)\right]$$

using property

$$A \cos(\omega_c t + \theta)$$

Amp — Freq — Phase

For carrier

In amplitude modulation Amplitude vary in proportion to $m(t)$

Amplitude Modulation is defined as a process in which the amplitude of the carrier wave is varied about a mean value Linearly with the baseband signal $m(t)$

A signal at higher frequency travel large distance
Carrier is always at higher frequency
So use to travel at larger distance.

We superimposed our original signal with carrier to travel larger distance

Demodulation :

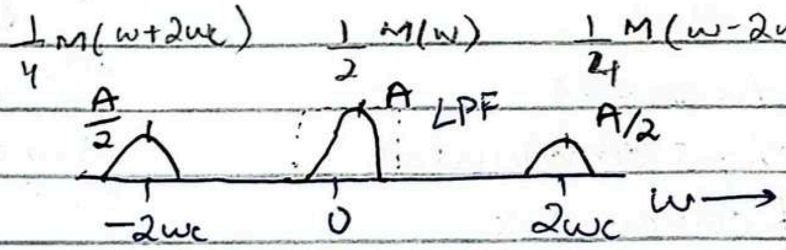
Demodulation consists of multiplication of incoming modulated signal $m(t)\cos\omega_c t$ by the carrier $\cos\omega_c t$

$$e(t) = m(t)\cos^2\omega_c t$$
$$= \frac{1}{2}\left[m(t) + m(t)\cos 2\omega_c t\right]$$

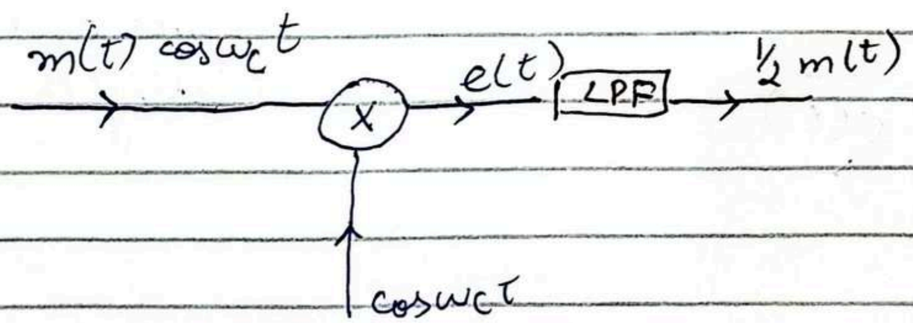
$$E(\omega) = \frac{1}{2}M(\omega) + \frac{1}{4}\left[M(\omega+2\omega_c) + M(\omega-2\omega_c)\right]$$

$\frac{1}{2}M(\omega) \rightarrow A$

$\frac{1}{4} \rightarrow \frac{A}{2}$



To recover $m(t)$ we need $M(\omega)$ and $M(\omega)$ is at center, which is LPF

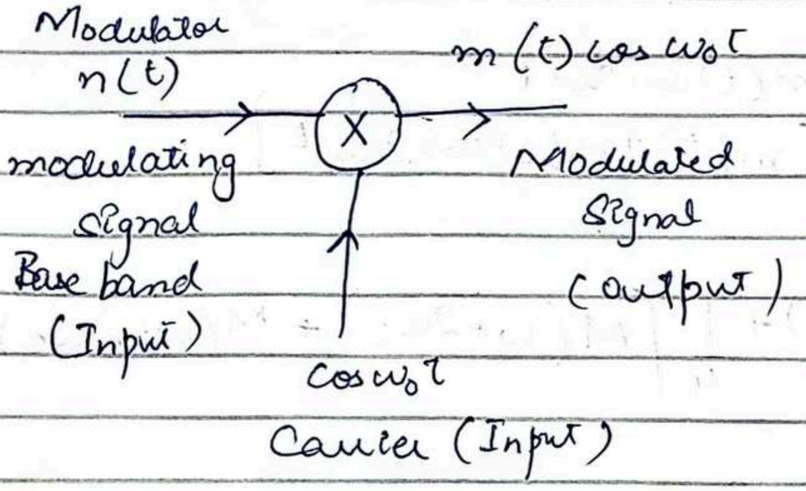


Demodulator

25 September

Lecture 11

Amplitude Modulation



Baseband signal superimpose carrier generate modulated signal

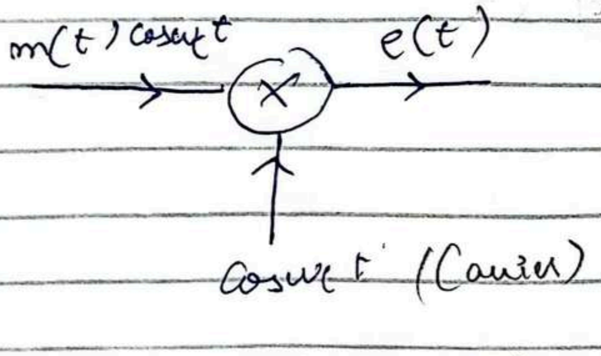
At this carrier at higher frequency can transmit to larger distance

Demodulator

↳ Synchronous
↳ Corehant demodulator
↳ Done at Reciver side

Synchronous
↓
bcz input & carrier have same frequency and phase

Principle of Synchronous
Same carrier as during transmission
we want to recive same carrier that should be in same frequency and phase



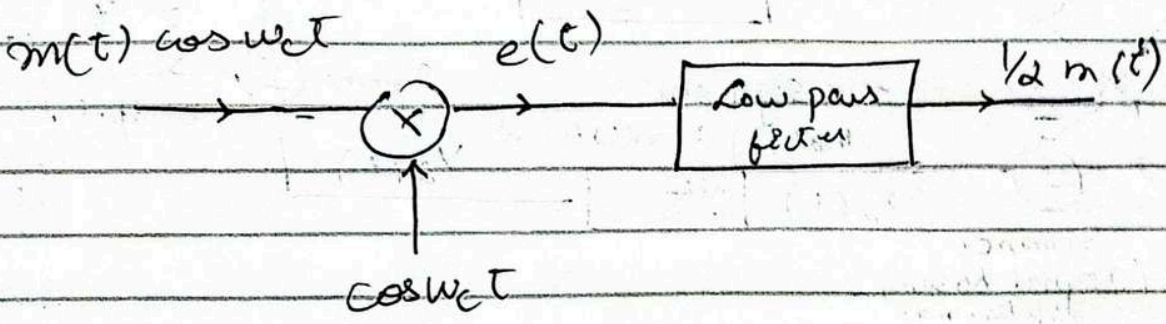
$$e(t) = m(t) \cos^2 \omega_c t$$

using trignometric identities

$$= \frac{1}{2}\left[m(t) + m(t) \cos 2\omega_c t\right]$$

$$e(t) = \frac{1}{2} m(t) + \frac{1}{2} m(t) \cos 2w_c t$$

we want to recover m(t) actual signal. we need m(t) which is baseband We add LPF to recover baseband (allow smaller frequ- to pass)



Types of Modulators

Multiplier Modulator

Analog multiplier with o/p proportional to the two i/p

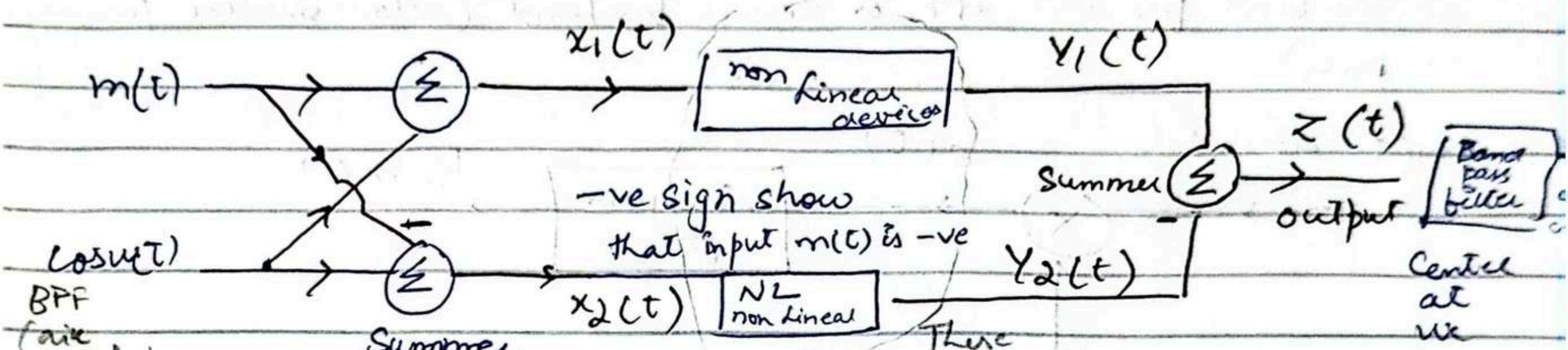
→ Difficult to maintain Linearity

CamScanner

eg. Output → product of 2 input.

Non Linear Modulator

(Single bandwidth modulator)



Summer (2 signal Ko sum) up kar raha

Balanced system

BPF (aik certain band ki frequency pass hogi)

LPF aik lipay nahi laghay ga bcz wo sari high frequencies ko pass kardy ga

$Y(t) = a\,x(t)$ Linear

$Y(t) = a\,x(t) + bx^2(t)$ non Linear

As system is non Linear so we used non Linear device (jis ki VI characteristics non Linear like transistor, diode)

$Y(t) = ax(t) + bx^2(t)$ (characteristics of non Linear dev

$$Z(t) = Y_1(t) - Y_2(t)$$
$$= \left[ax_1(t) + bx_1^2(t)\right] - \left[ax_2(t) + bx_2^2(t)\right]$$

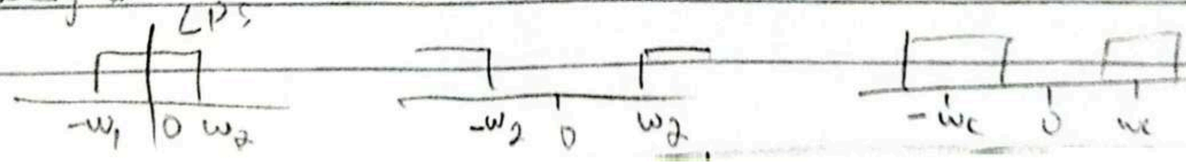
LPF nahi laghay ga bcz 1st Term a jahay ga

Put $x_1(t) = m(t) + \cos w_c t = \cos w_c t + m(t)$

$x_2(t) = -m(t) + \cos w_c t = \cos w_c t - m(t)$

wo nahi chahay bcz modulator ma output ma carrier chahay hai

$$Z(t) = \Big[\underbrace{2a\,m(t)}_{\text{input}} + \underbrace{4b\,m(t)\cos w_c t}_{\text{Output require (as output as carrier)}}\Big]$$



30 Sep 2024

Lecture 11

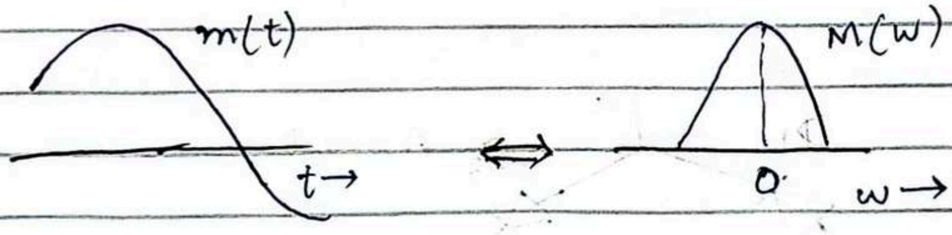
Signal goes through switching operation turn on & off periodically

Switching Modulation:

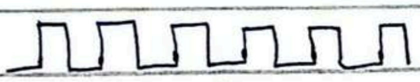
multiply signal with pulse train

Trying to perform switching operation periodic turn on & off the switch

Multiplication of a signal by a Square pulse train is in reality a switching operation which involves switching the signal $m(t)$ on and off periodically and can be accomplished by simple switching elements controlled by a pulse train $w(t)$

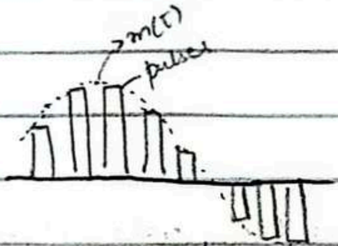


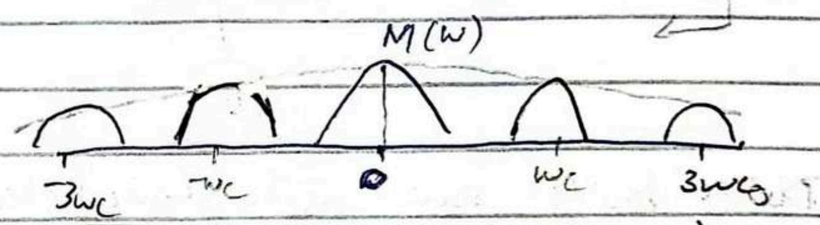
Spectrum keep getting smaller as we move away from center M(ω) strength becomes small



$w(t)$ is a periodic signal with Fourier Series

$$w(t) = \frac{1}{2} + \frac{2}{\pi}\left(\cos w_c t + \frac{1}{3}\cos 3w_c t + \frac{1}{5}\cos 5w_c t\right.$$

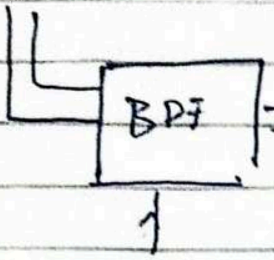




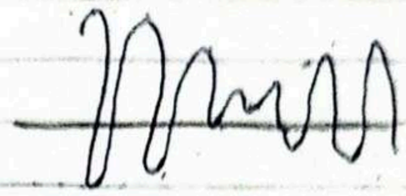
$$m(t)w(t) = \frac{1}{2}m(t) + \frac{2}{\pi}\left(\underbrace{m(t)\cos w_c t}_{\text{represent}} + \frac{1}{3}m(t)\cos 3w_c t + \cdots\right)$$

represent. we use modulation

This term is required as result of modulation we use band pass filter center at w_c



$= \frac{2}{\pi} m(t) \cos w_c t$

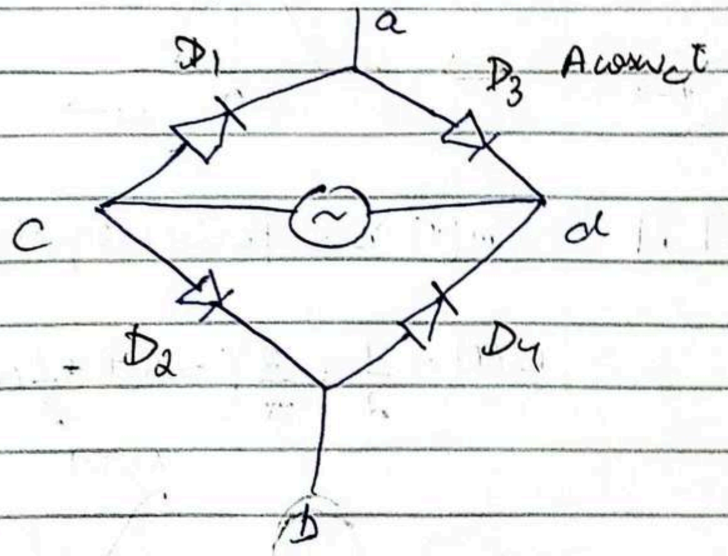


Centered at wc

$\frac{2}{\pi} m(t) \cos w_c t$

Acomplising Switching operation

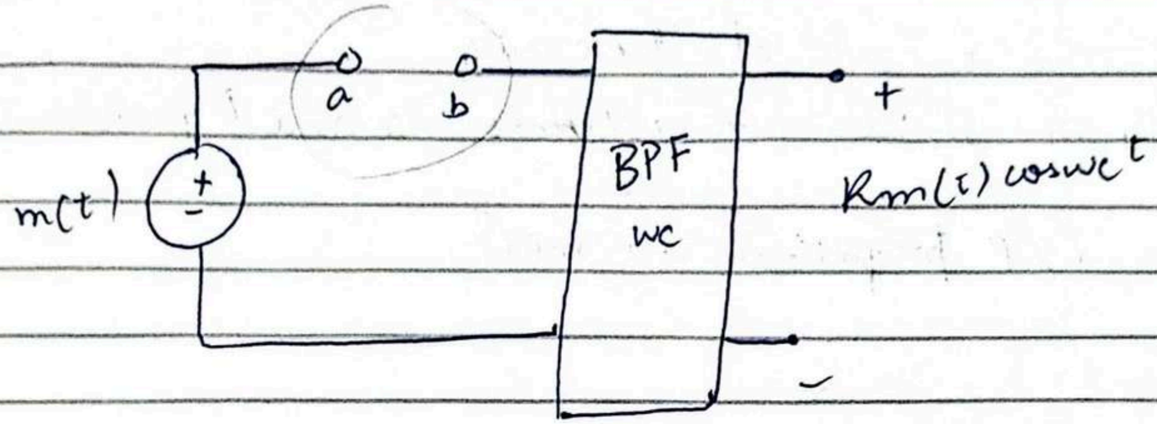
The switching operation can be achieved with a diode bridge modulator (which is the double balance modulator).



$f_c = \frac{w_c}{2\pi}$

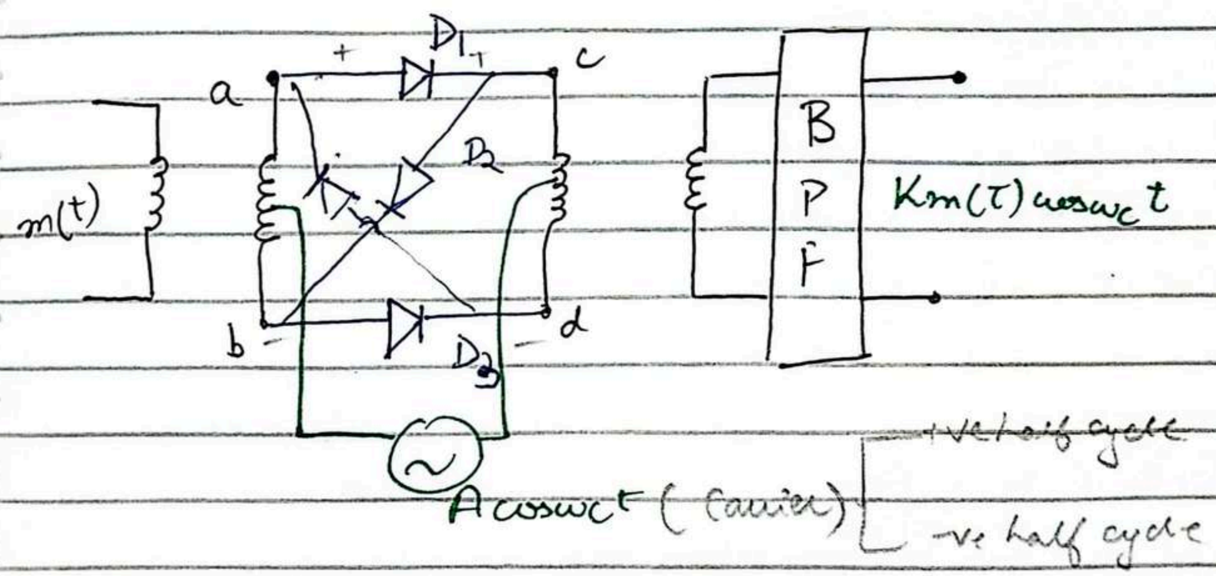
Diode bridge serves as the desired electronic switch when terminals a and b open and close periodically with carrier frequency fc when A coswct is applied across c d

Shows multiplying with $w(t)$



Full switching modulator which utilizes the diode bridge

Ring Modulator :



During the positive half cycle of carrier diodes D_1 and D_3 conducts whereas D_2 and D_4 are of

o/p proportional to $m(t)$

During -ve half cycle of the carrier

D_1 and D_3 are open whereas D_2 and D_4 conduct

O/P proportional to $-m(t)$

Amplitude Modulation → desire output coswct

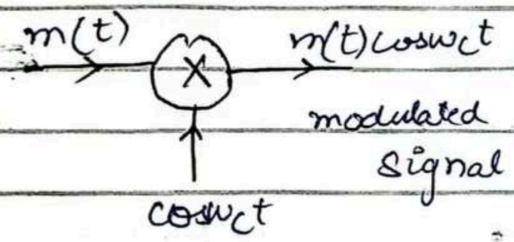
also switching modulator

During -ve half cycle of the carrier
D_1 and D_3 are open whereas D_2 and D_4 conduct

O/P proportional to $-m(T)$

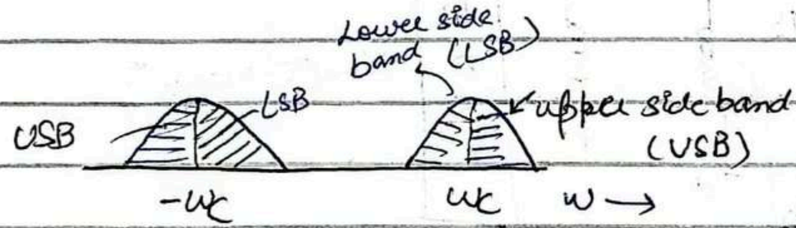
Lecture 13

Amplitude Modulation: (with carrier WC)



Message signal ⊗ with carrier signal general signal which is double side band Suppressed carrier

It can be both signal and double side



Double side band - suppressed carrier
(DSB-SC)

Characteristics of double side band.
↳ double side band
↳ Suppressed
↳ receiver costly hoga (bcz it is synchronous)
↳ generate same carrier again at recd

For Amplitude Modulation (AM)

SC (suppressed carrier)
(costly sophisticated receiver)
[Synchronous]
↳ Har receiver par synchronous demodulation karni parhay gi har signal ko apna carrier generate karna parhay ga

WC (with carrier)
[point to multipoint system]
↳ Example radio system
No synchronization need mean no carrier needed at receiver side

Amplitude Modulation with carrier

$$AM-WC \quad \phi_{AM}(t) = \underbrace{A\cos w_c t}_{\text{Carrier}} + \underbrace{m(t)\cos w_c t}_{\text{modulated signal}}$$ has carrier signal embedded

here we have 2 carrier signal input in original signal there is only one carrier signal

$$= [A + m(t)]\cos w_c t$$

if we take fourier transform

⇕

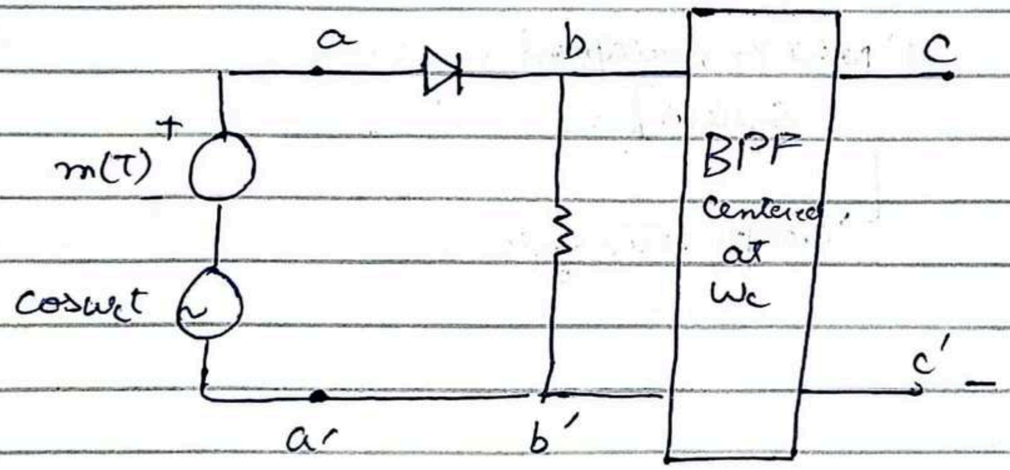
$$\phi_{AM}(w) = \frac{1}{2}[M(w-w_c)+M(w+w_c)]$$

$$+\pi A\underbrace{[\delta(w+w_c) + \delta(w-w_c)]}_{\text{Two additional pulse at } w_c \text{ and } -w_c}$$

high power conservation (disadvantage) more energy required

Generation of AM-WC Signal

AM signals can be generated by any DSB-SC modulator if the modulating signal is $[A+m(t)]$



Input is $c\cos\omega_c t + m(t)$ with $c \gg m(t)$ Diode act as a switching element and multiplies the signal $[c\cos\omega_c t + m(t)]$ with pulse train $w(t)$

$$V_{bb'}(t) = [c\cos\omega_c t + m(t)]\,w(t)$$

$$= [c\cos\omega_c t + m(t)]\left\{\frac{1}{2} + \frac{2}{\pi}\left(\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \frac{1}{5}\cos 5\omega_c t \ldots\right)\right\}$$

$$= \frac{c}{2}\cos\omega_c t + \frac{2}{\pi}m(t)\cos\omega_c t + \underbrace{\ldots\ldots}$$

$\frac{1}{2}m(t)$ is also suppressed

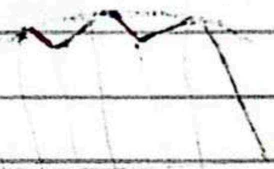
but here we picked terms at ω_c all other terms are suppressed

All of these terms are suppressed by BPF (ω_c)

constant does not matter carrier ... Modulated signal

constant does not matter

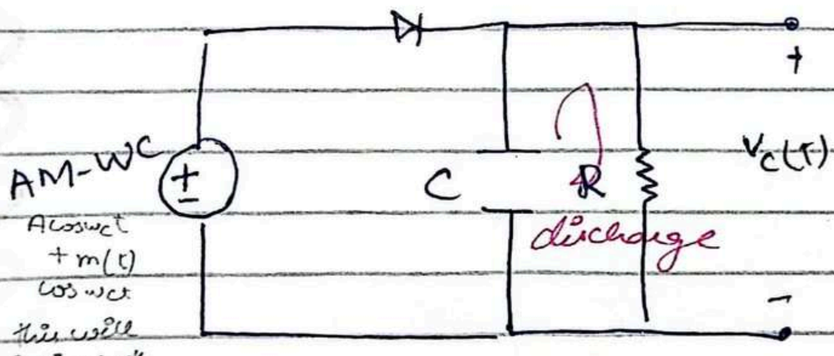
$$V_{cc}'(t) = \underbrace{\frac{C}{2}\cos\omega_c t + \frac{2}{\pi} m(t)\cos\omega_c t}_{AM-WC}$$



Benefit (cost effective receiver (where you donot have to achieve carrier signal again (synchronous))

Envelop Detector :

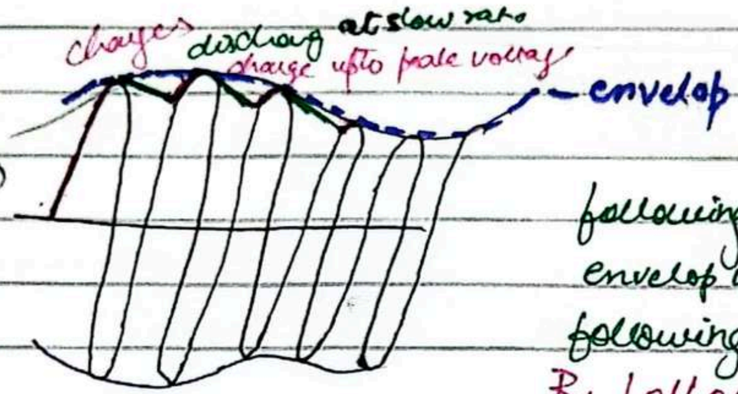
In an envelop detector, the output of the detector follows the enevelop of the modulated signal



$A\cos\omega_c t + m(t)\cos\omega_c t$

this will not work with AM only

On the positive cycle of input signal the diode conducts and capacitor seek charges upto the peak voltage of the input signal During negative half cycle the diode is open and capacitor discharges through the resistor R at a slow rate depending on the time constant RC The output voltage $v_c(t)$ closely follows the envelop of the input.



following the envelop of signal envelop was m(t) by closly following you will detect m(t)

By following envelop we are detecting the signal

Not Synchronous (no need carrier signal)

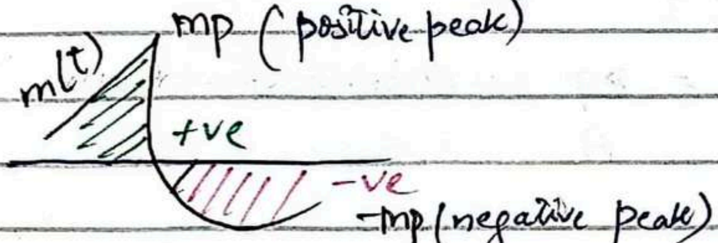
only capacitor, diode and resistor is attach (much simpler)

(with carrier transmission ki wajah sa receiver is simplified

Lecture 14

Broadcast → single transmitter (many receiver)

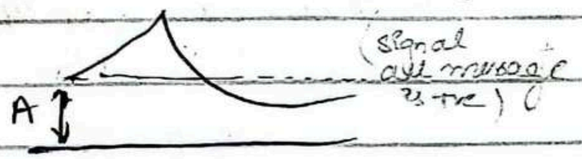
with carrier transmission → require double power



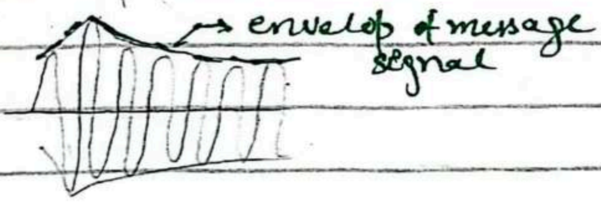
$\underbrace{\overbrace{A}^{\text{DC value}} + m(t)}_{\text{envelop}}$

Condition for envelop detection:

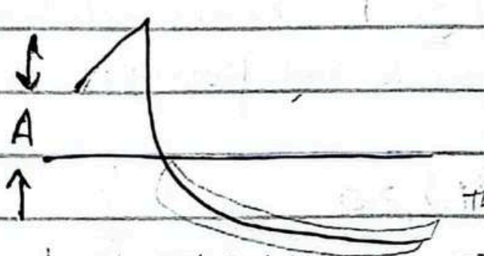
$A + m(t) > 0$ for all t
(A is taken very large)



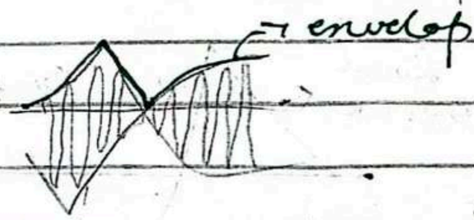
here envelop detection can take place



$A + m(t) \not> 0$
(A is taken smaller)



here envelop detection cannot take place but as carrier is transmitted as it is so coherent detection can be perform



Condition for envelop detection
$A + m(t) \geq 0$ for all t

$A + m(t) > 0$ for all t

modulation index $\mu = \frac{m_p}{A} = \frac{\text{message peak}}{\text{amplitude of carrier}}$

$A + m(t) \geq 0$ implies that $A \geq m_p$

for envelop detection to be possible

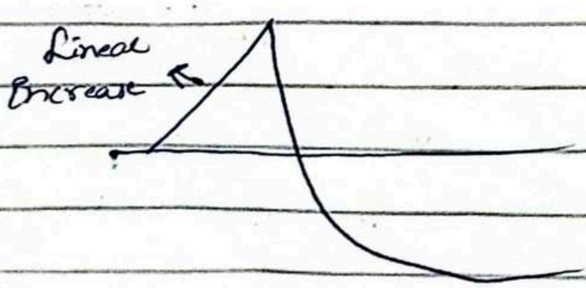
$\underbrace{0 \leq \mu}_{A > m_p} \leq 1$ (envelop detection can occur)

$A = m_p$

$A < m_p$ $\mu \geq 1$ overmodulation (and in this case envelop detection is not possible! (need to use synchronous demodulation/ coherent detection

Case 1 $m(t) \geq 0$

Case 2 $m(t) \ngeq 0$



If A is added than shifted A times but normal slope remain same

14 oct 2024

Lecture 15

Modulation Index $\mu = \frac{m_p}{A}$

→ ratio of message peak to the amplitude of carrier

max point where message signal reach in terms of amplitude

Tone modulation (carrier is also a sinusoid & message is also a sinusoid)

Example 4.4

in case of carrier w_c

Sketch $\varphi_{AM}(t)$ $\mu = 0.5$ and $\mu = 1$ when $m(t) = B\cos w_m t$ this case is refered to as tone modulation the modulating signal is a pure sinusoid (or tone)

$\mu = \frac{m_p}{A}$ → (bcz we want to define modulation index)

$\therefore m_p = B$ (only amplitude)

$\mu = \frac{B}{A}$

$B = \mu A$

As $m(t) = B\cos w_m t$

$m(t) = \mu A \cos w_m t$ $\quad (\therefore B = \mu A)$

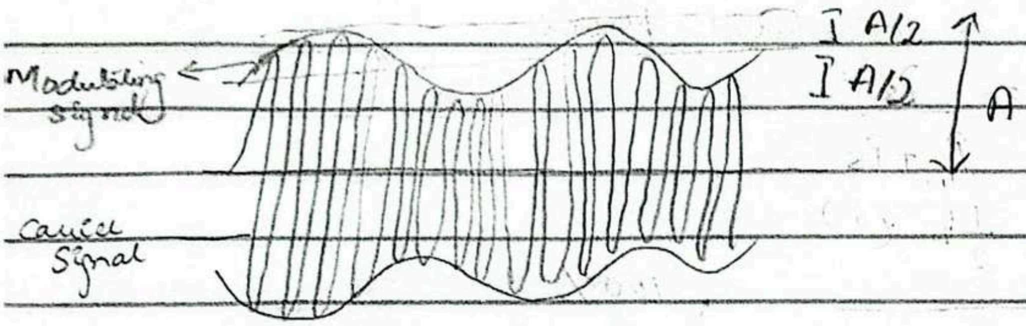
Therefore

$\varphi_{AM}(t) = [A + m(t)] \cos w_c t$

$= [A + \mu A \cos w_m t] \cos w_c t$

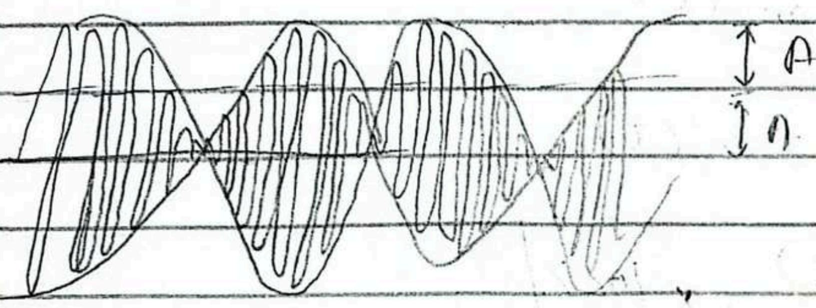
$= A[1 + \mu \cos w_m t] \cos w_c t$

$\mu = 0.5$



$(1+0.5)A$
when $\mu = 0.5$
* if $A = 1.5A$
then message peak
is at 1.5
* frequency of carrier
much higher than
frequency of message si

$\mu = 1$



When $\mu = 1$
$(1+1)A = 2A$
message peak $= 2$
if $\mu > 1$ than signal goes to
negative region so envelop det.
cannot take place (over mod

AM with carrier (difference power efficiency → power waste)

Sideband and Carrier power

$$\varphi_{AM}(t) = \underbrace{A \cos\omega_c t}_{\text{Carrier}} + \underbrace{m(t)\cos\omega_c t}_{\text{Sidebands}}$$

The carrier power P_C is the mean square value of $A\cos\omega_c t$

$$\boxed{P_C = \frac{A^2}{2}}$$

The side band power is the power of $m(t)\cos\omega_c t$ which is $0.5\,m^2(t)$ → need positive

$$\boxed{P_S = \frac{1}{2} m^2(t)}$$

(make sure that it remain Positive)

power efficiency $= \eta = \frac{\text{useful power}}{\text{total power}}$

$$= \frac{P_s}{P_c + P_s}$$

$$= \frac{\frac{1}{2}m^2(t)}{\frac{A^2}{2} + \frac{1}{2}m^2(t)} \times 100\%$$

$$= \frac{\cancel{\frac{1}{2}}(m^2(t))}{\cancel{\frac{1}{2}}(A^2 + m^2(t))}$$

$$\boxed{\eta = \frac{m^2(t)}{A^2 + m^2(t)} \times 100\%}$$

$\frac{m^2(t)}{A^2 + m^2(t)} \ngtr 1$

→ In this case efficieny $\approx 100\%$ if it be $= 1$ that mean efficieny 100%

→ In this $A = 0$ this mean there is no carrier signal

AM with carrier efficieny cannot be exceed than 25% (this is highly inefficient signal as less efficiency)

Example 4.5 Determine η and the percentage of the total power carried by the sideband of the AM wave for tone modulation when a) $\mu = 0.5$ b) $\mu = 0.3$

$\mu = \frac{B}{A} \Rightarrow B = \mu A$

For tone modulation $m(t) = \mu A \cos \omega_m t$

$$\overline{m^2(t)} = \frac{\mu^2 A^2}{2}$$

$$P_s = \frac{m^2(t)}{2} = \frac{\frac{(\mu A)^2}{2}}{2}$$

$$P_c = \frac{A^2}{2}$$

$$\eta = \frac{P_S}{P_C + P_S} = \frac{\left(\frac{\mu A}{2}\right)^2 / 2}{\frac{A^2}{2} + \left(\frac{\mu A}{2}\right)^2 / 2}$$

$$= \frac{\frac{\mu^2 A^2}{4}}{\frac{A^2}{2} + \frac{\mu^2 A^2}{4}}$$

$$= \frac{\mu^2 A^2}{2A^2 + \mu^2 A^2}$$

$$= \frac{\cancel{A^2}(\mu^2)}{\cancel{A^2}(2 + \mu^2)} = \frac{\mu^2}{2 + \mu^2}$$

for $\mu = 0.5$

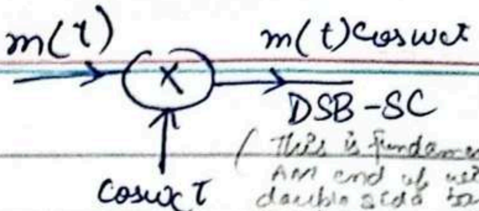
$$\eta = \frac{(0.5)^2}{2 + (0.5)^2} \times 100\%$$

$\boxed{\eta = 11.1\%}$ only 11% of total power is useful

for $\mu = 0.3$

$$\eta = \frac{(0.3)^2}{2 + (0.3)^2} \times 100\%$$

$\boxed{\eta = 4.3\%}$



(This is fundamental AM and of with double side band and always give out can)

Example 4.1 (Tone modulation)

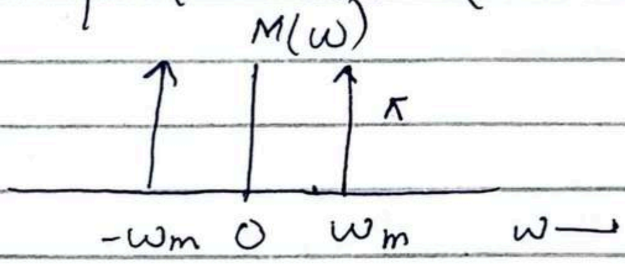
baseband signal $m(t) = \cos w_m t$ find DSB-SC and sketch its spectrum identify USB and LSB verify that DSB SC modulated signal can be demodulated by the demodulator

$w = 2\pi f$

$$m(t) = \cos w_m t$$

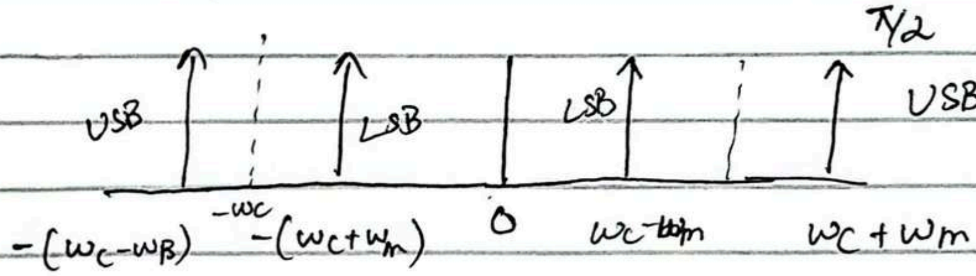
$$M(\omega) = \pi\left[\delta(\omega - \omega_m) + \delta(\omega + \omega_m)\right]$$

[cosine gives fourier transform 2 impulses]



Spectrum are 2 impulses

Spectrum consist of two impulses at ± Wc



(Suppressed Carrier)

spectrum beyond w_c is the USB and the one below w_c is the LSB

In the time domain approach we work directly with signals in time domain for baseband signal $m(t) = \cos w_m T$

$$= \cos w_m t \cos w_c t$$

$$= \frac{1}{2}\left[\cos(w_m + w_c) - \cos(w_m - w_c)\right]$$

16 Oct 2026

Lecture 16

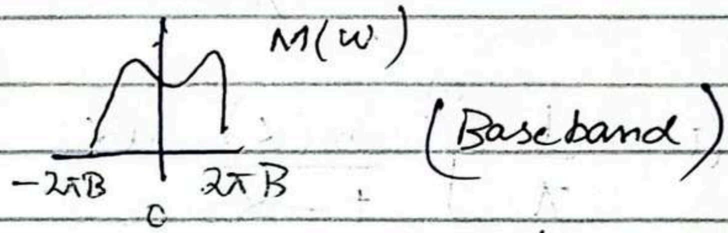
$m(t) \cos \omega_c t$

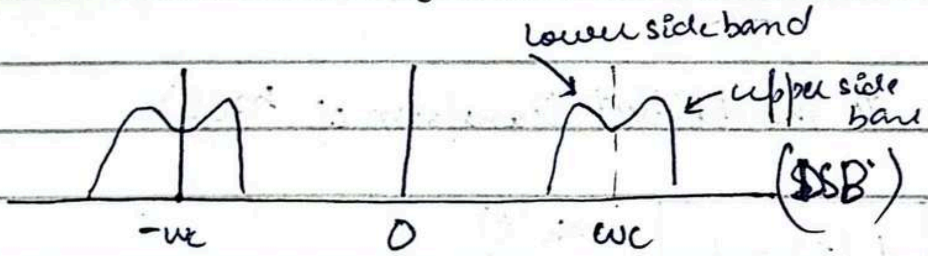
DSB – SC

DSB, SSB (LSB, USB)

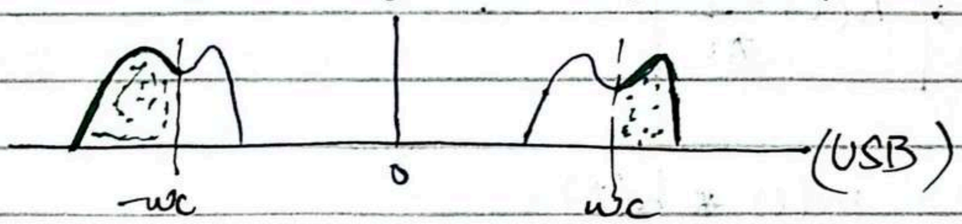
WC, SC

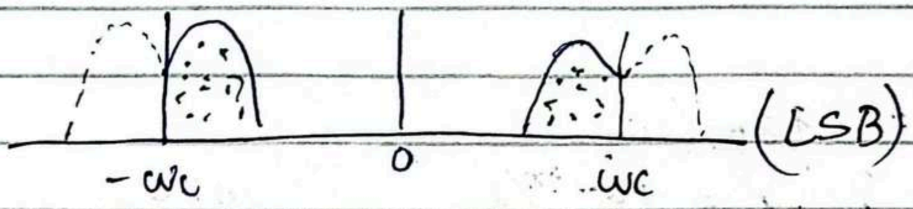
Single side band (reduce bandwidth)

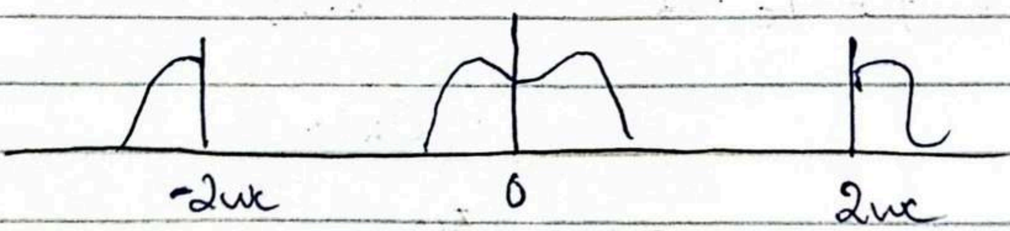




Coherent
↓
multiplication with carrier







Hilbert Transform:

$m(t) \qquad m_h(t)$

$$m_h(t) = m(t) * \frac{1}{\pi t}$$

$$m_h(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{m(\alpha)}{t-\alpha} \, d\alpha$$

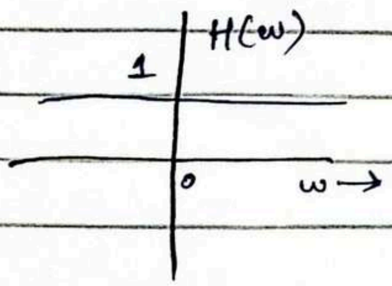
$m_h(t)$ is the hilbert transform of $m(t)$

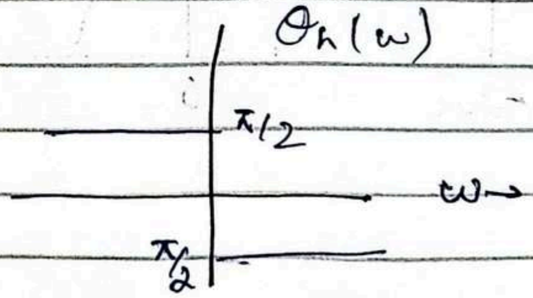
$$m(t) \longrightarrow \boxed{h(t) = \frac{1}{\pi t}} \longrightarrow m_h(t)$$

$$m_h(t) = m(t) * h(t)$$

$$H(\omega) = -j\,sgn(\omega)$$

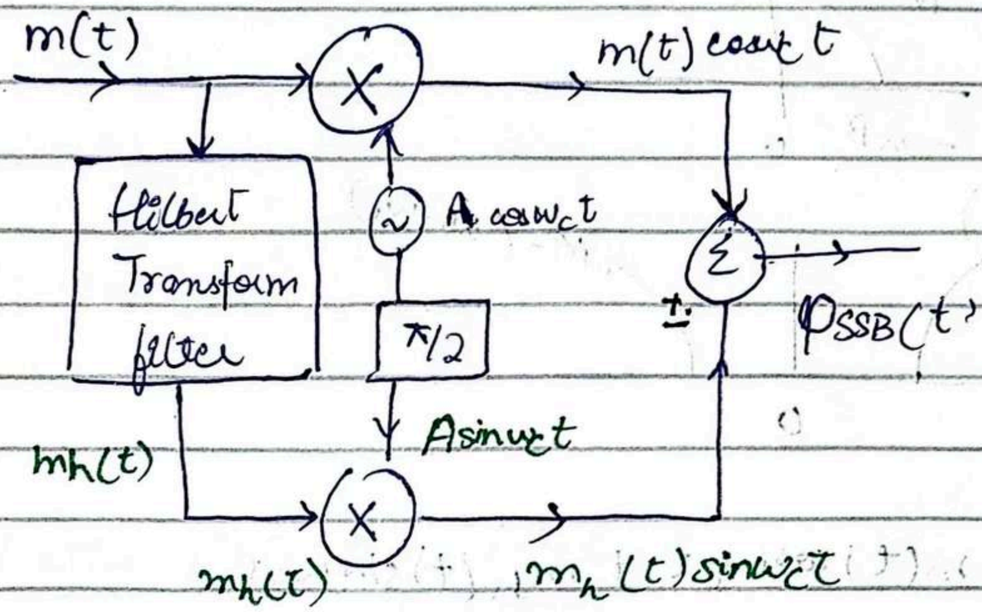
$$= \begin{cases} -j = 1e^{-j\pi/2} & \omega > 0 \\ j = 1e^{j\pi/2} & \omega < 0 \end{cases}$$



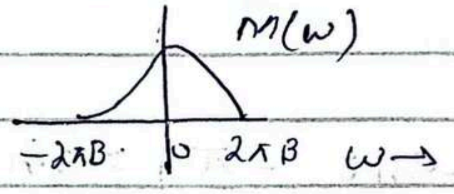


Phase shift signal $m(t)$ by $\pi/2$

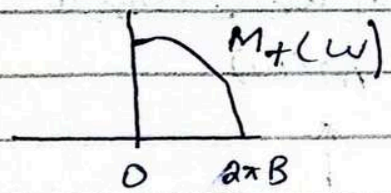
Generation of SSB using Hilbert Transform

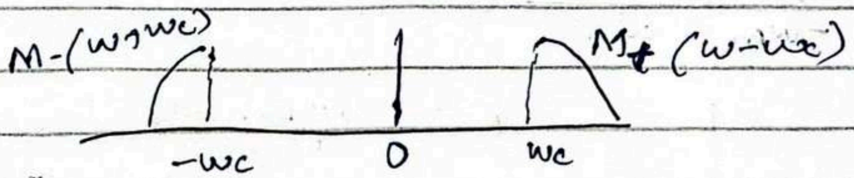


$$\varphi_{SSB}(t) = m(t)\cos\omega_c t \pm m_h(t)\sin\omega_c t$$



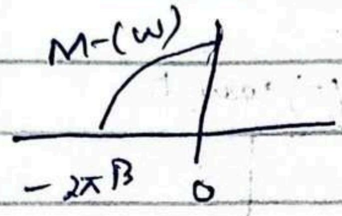
USB $M_+(\omega) = M(\omega)u(\omega)$

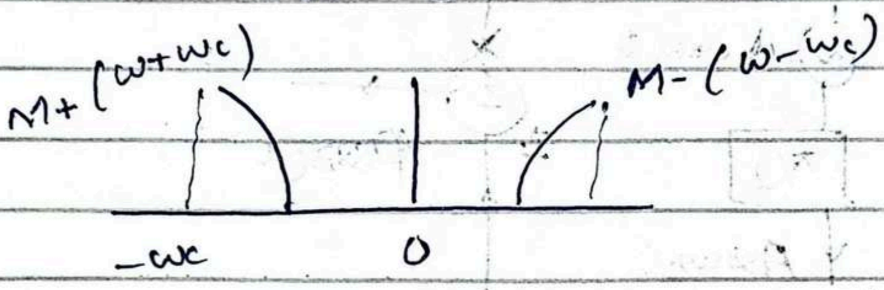




$$\varphi_{USB}(t) = m(t)\cos\omega_c t - m_h(t)\sin\omega_c t$$

LSB $M_{-}(\omega) = M(\omega)\,U(-\omega)$





$$\varphi_{LSB}(t) = m(t)\cos\omega_c t + m_h(t)\sin\omega_c t$$

$$\varphi_{SSB}(t) = m(t)\cos\omega_c t \mp m_h(t)\sin\omega_c t$$

Example 4.7 Tone Modulation Signal

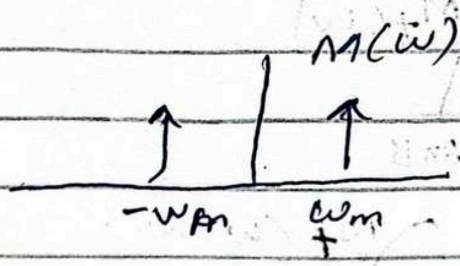
$m(t) = \cos\omega_m(t)$

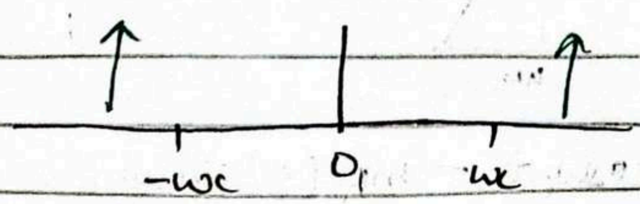
Find $\varphi_{SSB}(t)$

$$m_h(t) = \cos\left(\omega_m t - \frac{\pi}{2}\right) = \sin\omega_m t$$

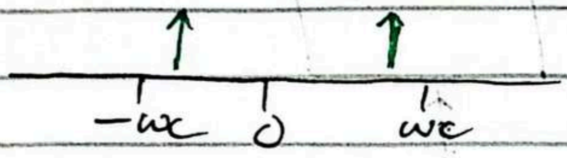
$$\varphi_{SSB}(t) = \cos\omega_m t\cos\omega_c t \mp \sin\omega_m t\sin\omega_c t$$

$$= \cos(\omega_c \mp \omega_m)t$$

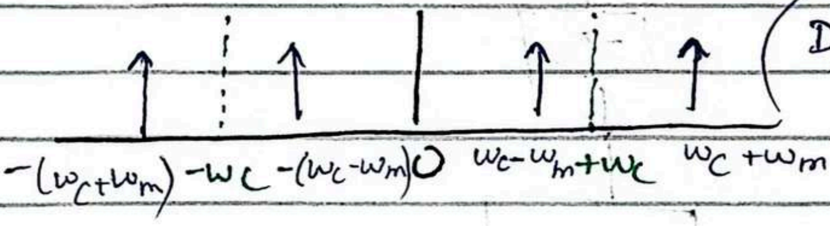




Upper side band
(Choose +ωc)
b/w



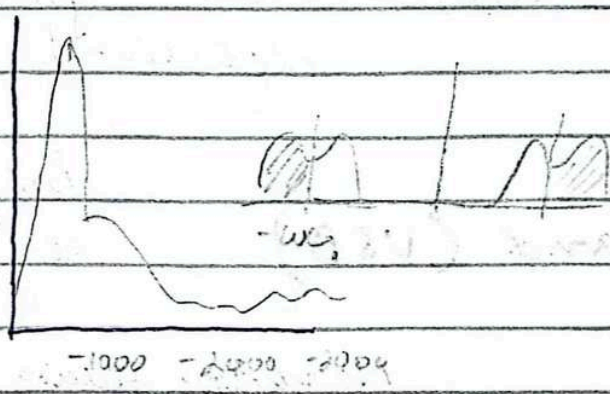
(LSB) (-ve sign)



(DSB spectrum)

(error as we require SSB but it shows both DSB)

Selective Filtering Method (method of generating SSB)



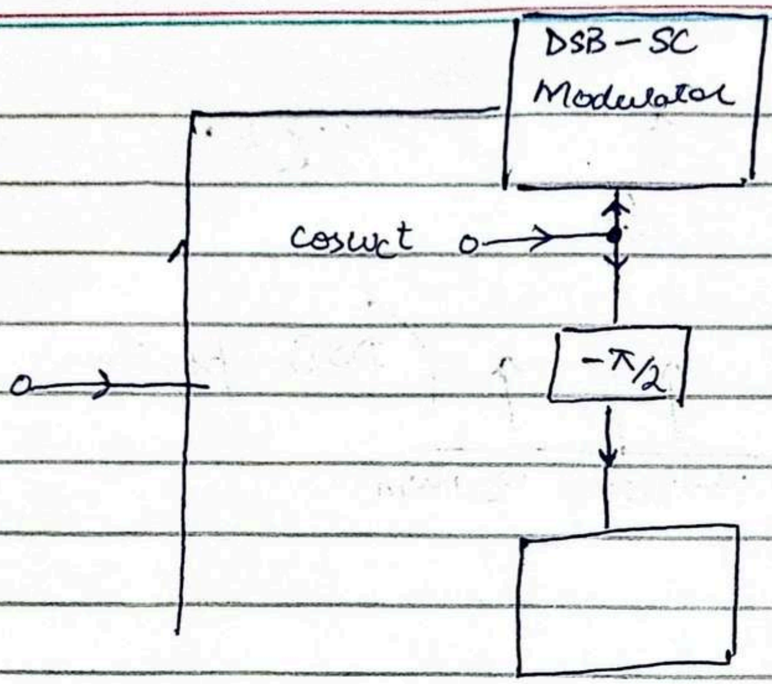
→ filter axis is at ω_c

→ it is not bandpass Low pass high pass

→ it filter after ω_c like only uppersede band

Phase Shift Method

$$\varphi_{SSB}(t) = m(t)\cos\omega_c t \mp m_h(t)\sin\omega_c t$$

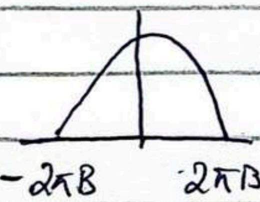


Suppressed Carrier (need another carrier → Synchronous detection)

Brick wall
↓
Straight side

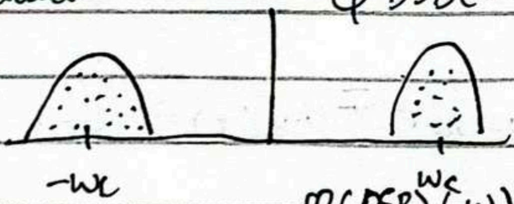
AM : Vestigial Side Band (VSB)

100% Bandwidth

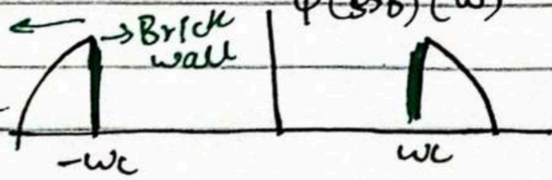


No selective filtering
Gradual filtering done

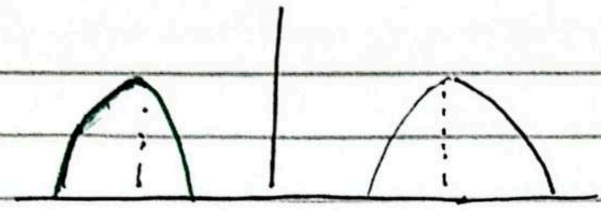
200% bandwidth



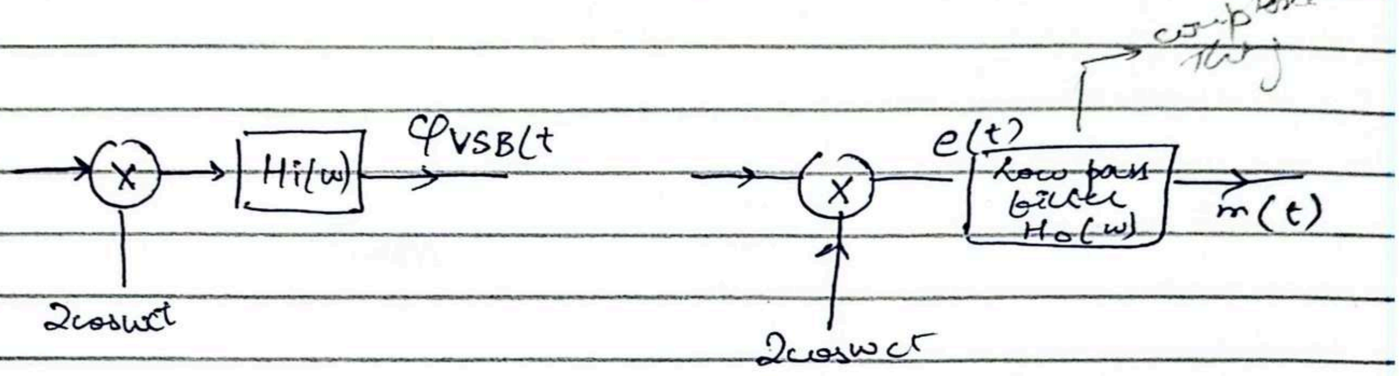
Steep cut-off
100% bandwidth



25% extra



Gradually cut off



Transmitter   Receiver

Hi(w) is VSB Shaping filter

$$e(t) = 2\varphi_{VSB}(t)\cos w_c t \Leftrightarrow \varphi_{VSB}(w+w_c) + \varphi_{VSB}(w-w_c)$$

$$M(w) = [\varphi_{VSB}(w+w_c) + \varphi_{VSB}(w-w_c)]\, H_o(w)$$

$$M(w) = M(w)[H_i(w+w_c) + H_i(w-w_c)]\, H_o(w)$$

$$H(w) = \frac{1}{H_i(w+w_c) + H_i(w-w_c)} \qquad |w| \le 2\pi B$$

21 oct 2024

Lecture 17

Chapter 5

Angle Modulation

Technique of modulation where the angle of the carrier is varied with a modulating signal $m(t)$ are known as angle modulation. Two possibilities are phase modulation (PM) and frequency modulation (FM)

$$\varphi_{PM}(t) = A\cos[\omega_c t + K_p m(t)]$$ (Phase modulation)

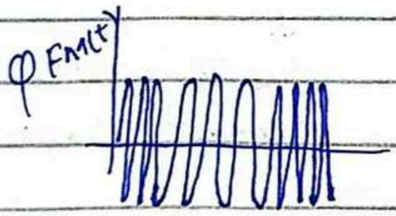
$$\varphi_{FM}(t) = A\cos\left[\omega_c t + K_f \int_{-\infty}^{t} m(\alpha)\, d\alpha\right]$$ (frequency modulation)

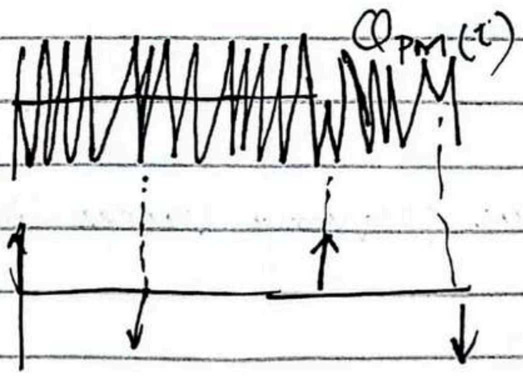
ω_i – Instantaneous frequency

Generalized sinusoid $A(\cos\omega_c t + \theta_0)$

↑ Angle of sinusoid

This angle actually vary in phase modulation





Frequency of carrier changes with carrier signal

Instantaneous Frequency

$$\phi(t) = A\cos(\underbrace{\omega_c t}_{\text{central frequency}} + \underbrace{\theta_c}_{\text{angle}})$$

The instantaneous frequency ω_i at any instant t is the slope of $\theta(t)$ at t. Thus for $\phi(t)$

Instantaneous frequency ← $\boxed{\omega_i(t) = \frac{d\theta}{dt}}$ ←

↓

frequency at a particular 't' (that can change)

$$\theta(t) = \int_{-\infty}^{t} \omega_i(\alpha)\,d\alpha$$ (change in angle allow angle modulation)

The possibility of transmitting the information of $m(t)$ by varying angle θ of the carrier arises

$$\phi_{PM}(t) = A\cos\left[\omega_c t + k_p\, m(t)\right]$$

$$\phi_{FM}(t) = A\cos\left[\omega_c t + k_f \underbrace{\int_{-\infty}^{t} m(\alpha)\,d\alpha}_{\theta(t)}\right]$$

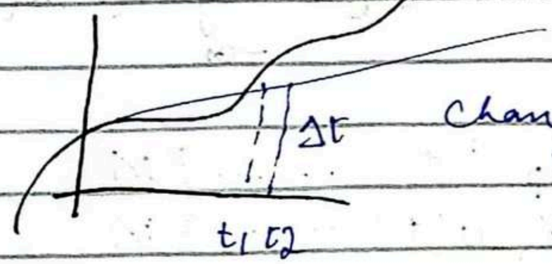
$\theta(t)$ ↓ due to which linearly changing

Everything we are talking about base on ω_i

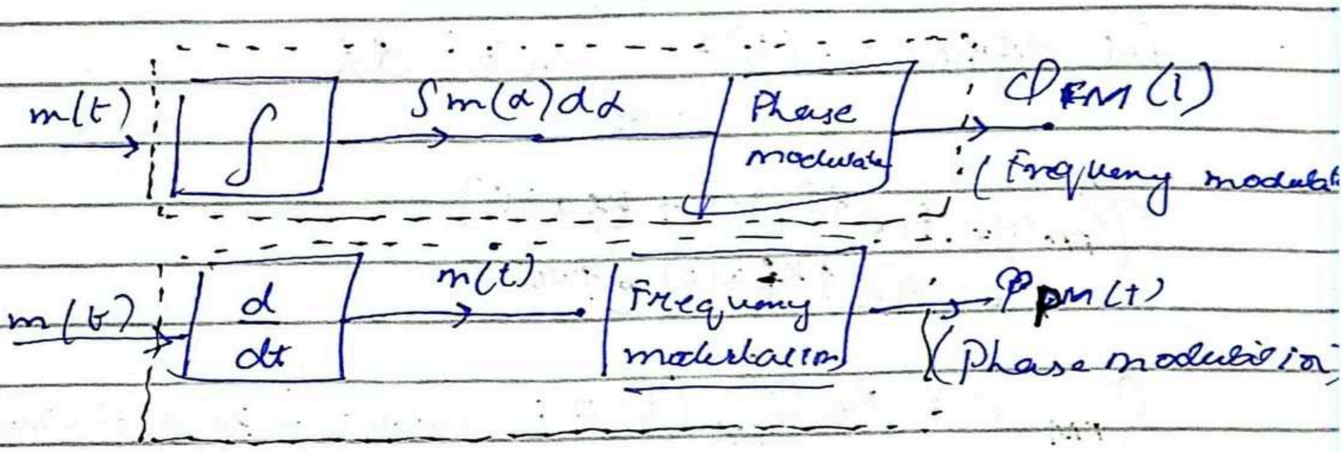
Instantaneous frequency varies linearly with the modulating signal in FM

$\varphi(t) = A\cos(\omega_c t + \theta_0)$ $\quad t_1 < t < t_2$

Signal carrier change of information through changes in time
Change in frequency is carrying the change in information



Changes in instantaneous angle



$$\varphi_{PM}(t) = A\cos[\omega_c t + K_p\, m(t)]$$

$$\varphi_{FM}(t) = A\cos\left(\omega_c t + K_f \int_{-\infty}^{t} m(\alpha)\, d\alpha\right]$$

dummy variable

Phase Modulation (Amplitude and frequency remain same angle changes)

Bandwidth of angle modulated signal

Generale angle modulated carrier

$$\varphi_{EM}(t) = A\cos[w_c t + \psi(t)]$$

↳ Electro magnetic waves ↳ Same as θ

$$= A\cos\left[w_c t + \int_{-\infty}^{t} m(\alpha)h(t-\alpha)\,d\alpha\right]$$

If $h(t) = k_p\,\delta(t)$ ...... PM

If $h(t) = K_f\,u(t)$ ..... FM

Let define $a(t) = \int_{-\infty}^{t} m(\alpha)\,d\alpha$

power series expansion

Bandwidth requirement for real theoratical become infinite

$$\hat{\varphi}_{FM}(t) = Ae^{j[w_c t + K_f a(t)]}$$

$$= Ae^{jK_f a(t)}\,e^{jw_c t}$$

$$\varphi_{FM}(t) = \mathrm{Re}\,\hat{\varphi}_{FM}(t) = A\Big[\cos w_c t - K_f\,a(t)\sin w_c t$$

$$- \frac{K_f^2}{2!}a^2(t)\cos w_c t + \frac{K_f^3}{3!}a^3(t)\sin w_c t + \cdots\Big]$$

These terms become very small. so we can ignore them but this is in narrow band

Theoratical BW of an FM wave is infinite

Narrow Band FM

Condition

$|K_f\,a(t)| \ll 1$

$$\varphi_{FM}(t) \simeq A[\cos w_c t - K_f\,a(t)\sin w_c t]$$

28 oct 2024

Lecture

Angle Modulation

FM

PM

Instantaneous Frequency

BW of FM waves

NBFM

WBFM

Evaluated certain values for FM

FM Modulation and Demodulation

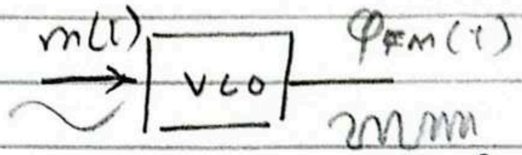
Frequency modulators are [Time varying / non Linear systems

Oscillators (Voltage controlled oscillator VCO) are designed to produce Frequency modulators

Oscillator design — Either Reactance Tube — or Varactor Diode

Varactor Diode → Capacitor whose capacitance changes with applied voltage

Direct FM Modulation : VCO



Voltage control oscillator (VCOs) are used to vary the frequency the carrier signal in accordance with the baseband signal amplitude variations (most used method)

Hartley or Collpitt Oscillator:

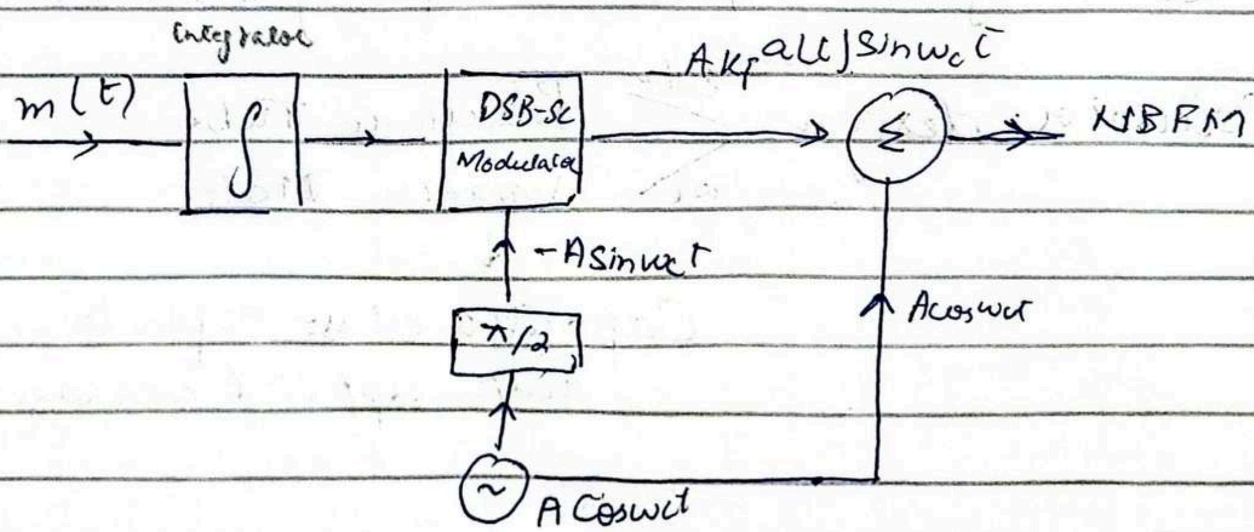
Frequency of oscillator = $\omega_0 = \dfrac{1}{\sqrt{LC}}$

→ In VCOs : W_0 varies linearly with the controlled voltage

→ VCOs can be controlled using OP amp and a hysteresis comparator (such as Schmitt Triger)

→ Reactive parameter (C or L) of the resonant oscillator can be varied to achieve VCO functionality

Narrow band FM wave generation



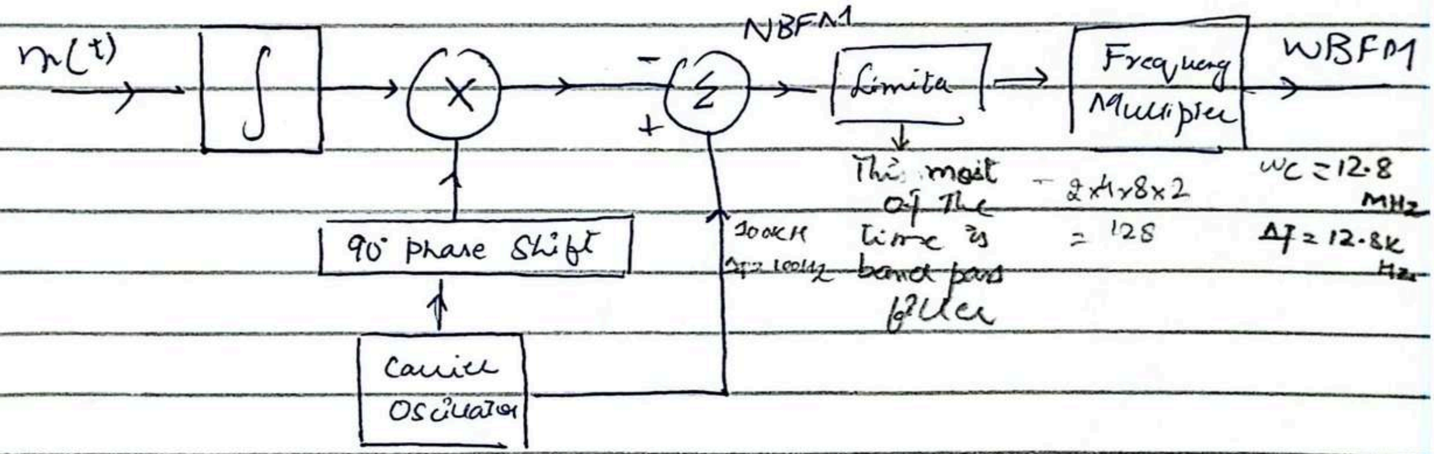
$\Delta f \rightarrow$ frequency deviation

If $\Delta f = 0$ so it is no FM signal
it cannot be possible

$$\underset{NBFM}{\varphi_{FM}(t)} \simeq A\left[\cos\omega_c t - k_f\, a(t) \sin\omega_c t\right]$$

convert NBFM to WBFM

Armistrong indirect method of FM Generation



A Wideband FM signal is produced by multiplying in frequency the narrowband FM signals using frequency multiplier

Lecture 20

Demodulation of FM

Information in an FM signal resides in the instantaneous frequency

$$\omega_i = \omega_c + K_f\, m(t)$$

So

$|H(\omega)| = a\omega + b$ over the FM band would yield output proportional to instantaneous frequency directly

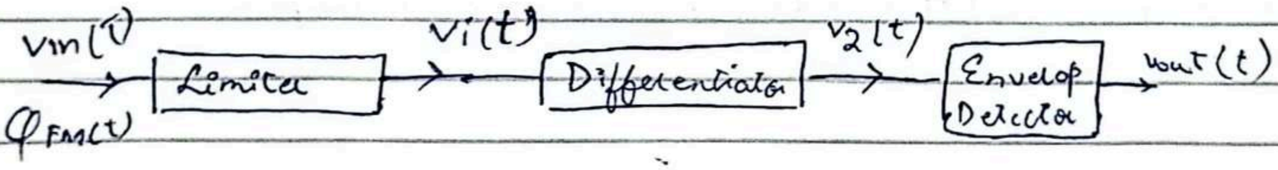
"Frequency to amplitude conversion circuit".

1. Slope detector: Mathematical analysis

2. Zero Crossing Detector ~~G/a~~ Graphical Analysis

3. Phase Locked Loop PLL

Slope Detector



$$v_i(t) = A\cos(2\pi f_c t + \theta(t))$$

↳ if there is no $\theta(t)$ than there is no angle and frequency modulation

$$= A\cos\left(2\pi f_c t + 2\pi K_f \int_{-\infty}^{t} m(\alpha)\, d\alpha\right)$$

After differentiation

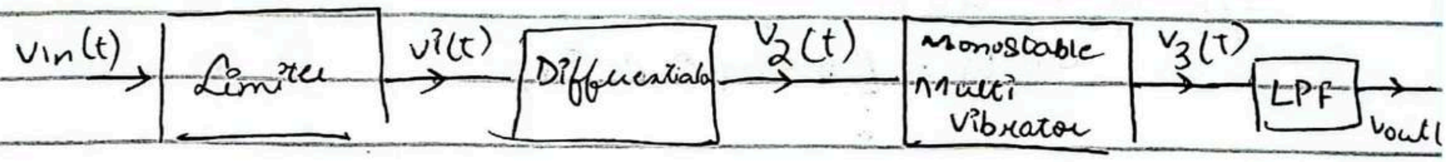
$$V_2(t) = \frac{d}{dt} V_1(t) = -A\underbrace{\left[2\pi f_c + \frac{d\theta}{dt}\right]}_{\text{amplitude of Sinusoid}} \underbrace{\sin(2\pi f_c t + \theta(t)}_{\text{Sinusoid}}$$

FM demodulation can be performed by taking the time derivative often call slope detector of FM signal followed by the envelope detection

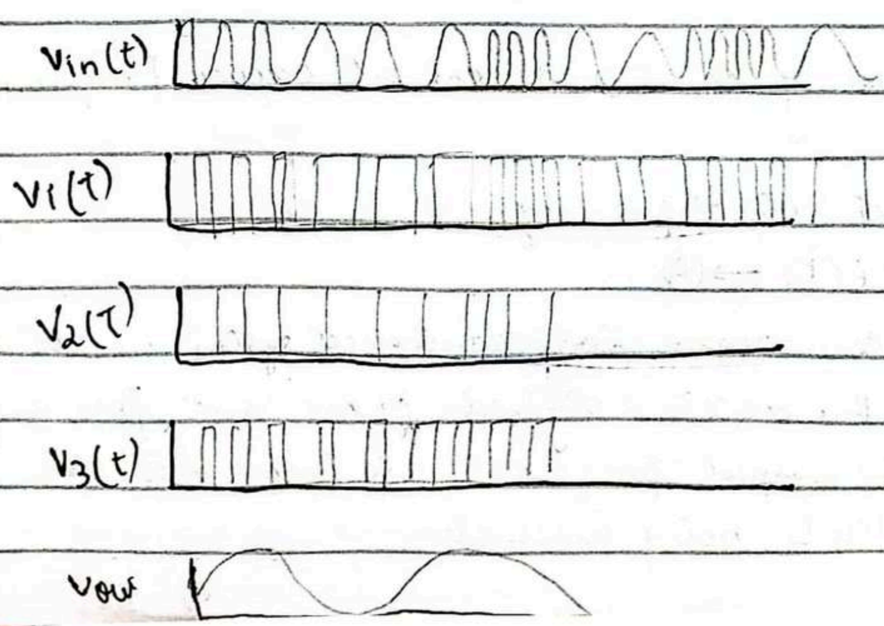
$$V_{out}(t) = A\left[2\pi f_c + \frac{d\theta}{dt}\right]$$

$$V_{out}(t) = A 2\pi f_c + A 2\pi K_f m(t)$$

Zero crossing detector



Frequency to amplitude conversion



after passing through mono stable vibrator stable pulse is produced

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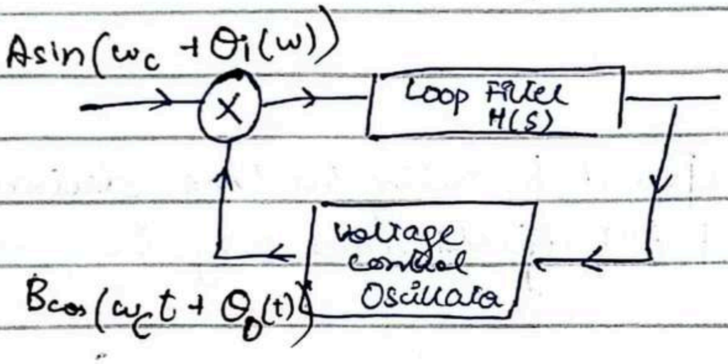
Lecture 21

FM Demodulation using Phase Locked Loop



Multiplier output is

$$AB \sin(w_c t + \theta_i) \cos(w_c t + \theta_o)$$

$$\frac{AB}{2}\left[\sin(\theta_i - \theta_o) + \sin(2w_c + \theta_i - \theta_o)\right]$$

Loop filter $H(s)$ being a low pass filter (LPF) supress the 'sum frequency term'

I/P to the VCO is

$$e_o(t) = \frac{AB}{2}\sin(\theta_i - \theta_o)$$

The free running frequency of VCO is set at carrier frequency w_c

Instantaneous frequency of VCO is given by

$$w_{co} = w_c + Ce_i(t) \rightarrow (4)$$

The phase locked loop is a closed loop controlled system, which can tracked variation in the received signal phase and frequency. In the lock state, the VCO output frequency is lock onto the frequency of input signal being tracked

Lock State
↳ Capture range
↳ Locked range

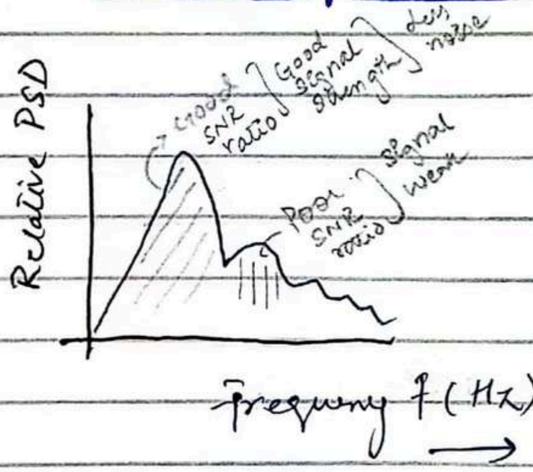
Once the VCO frequency is locked to the input frequency the output of the PLL is simply the distorted version of $m(t)$

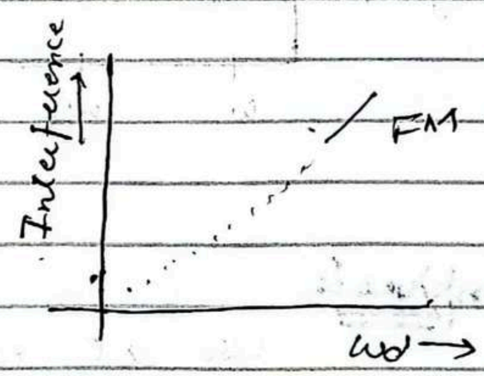
If VCO o/p is $B\cos(w_c t + \theta_o)$
than its instantaneous frequency is $w_c + \dot{\theta}_o(t)$ and comparing with (A)

$$\dot{\theta}_o(t) = c e_o(t)$$

$e_o(t)$ is the $m(t)$ when the VCO is locked to the input signal

Pre Emphasis & De Emphasis in FM system





Less than 2KHz signal to noise ratio Good
2KHz – 4KHz : signal to noise ratio poor

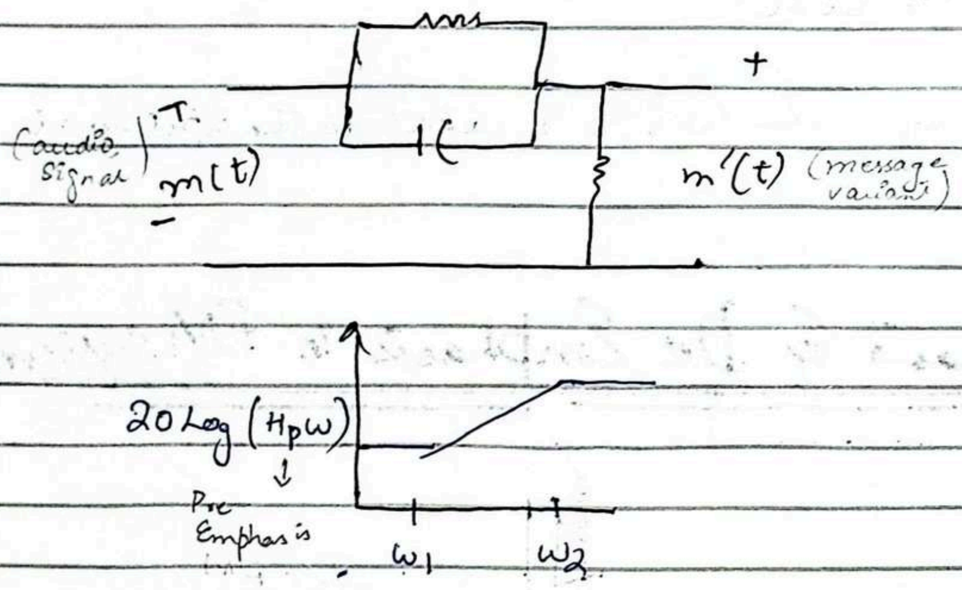
The interference of noise increases linearly with frequency in FM system and thus the noise power in receiver output is concentrated at higher frequency. The power spectral density of an audio signal $m(t)$ is concentrated at lower frequencies below 2.1KHz

Thus the weak $m(t)$ value the noise power is concentrated.

Pre Emphasis and De Emphasis filter

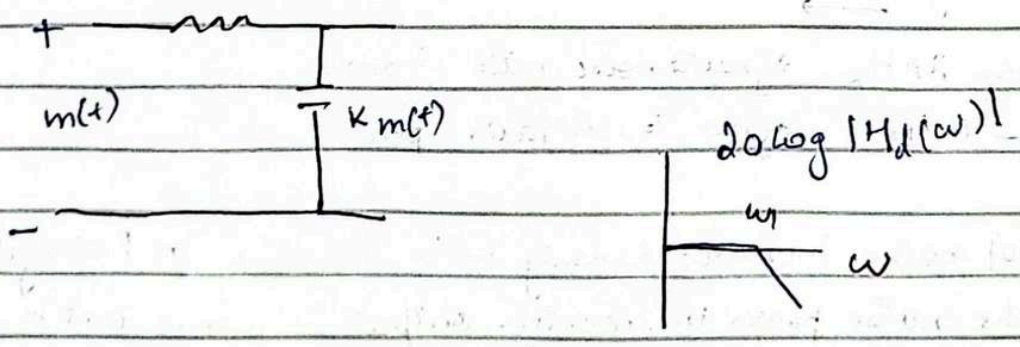
Pre Emphasis

use before modulation at transmitter side

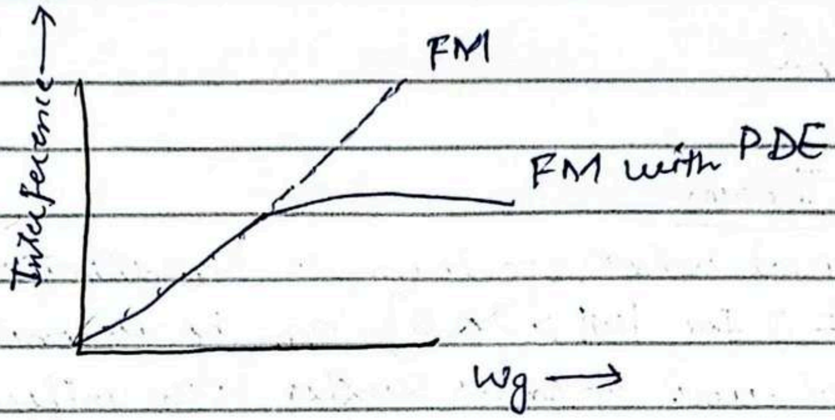


De Emphasis

use at receiver side



Final



The weaker high frequency components beyond 2.1KHz of audio signal m(t) are boosted before modulation by a pre Emphasis filter

F_1 corresponding to ω_1 is chosen as 2.1KHz and f_2 is typically 30KHz or more which is well beyond the audio range. The noise signal at higher frequencies get de emphasis

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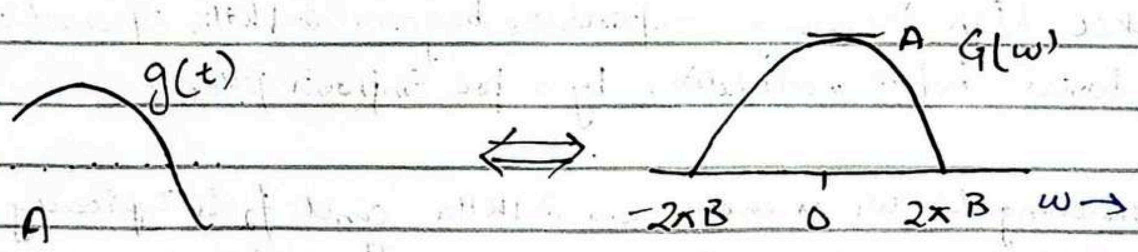
Lecture 22

Sampling Theorem (done on x axis)

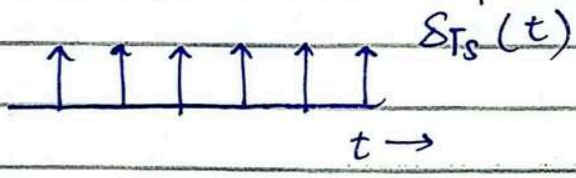
mean spectrum beyond B cannot achieve

Any signal whose spectrum is bandlimited to B Hz [$G(\omega) = 0$ for $|\omega| > 2\pi B$] can be reconstructed exactly (without error) from its samples taken uniformly at a rate $R > 2B$ samples/second



Assumption $g(t)$ is band limited to "B" Hz

Sampling Multiplication by impulse train



$$\delta_{T_s}(t) = \frac{1}{T_s}\left[1 + 2\cos\omega_s t + 2\cos 2\omega_s t + 2\cos 3\omega_s t + \cdots\right]$$

$$\boxed{\omega_s = \frac{2\pi}{T_s} = 2\pi f_s}$$

time ↔ sampling

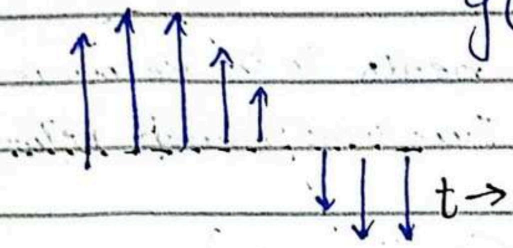
$T_s \rightarrow$ Sampling Interval

$f_s \rightarrow$ Sampling frequency

Maximum sampling frequency

$$f_{s_{min}} = 2B\,Hz$$

$$\bar{g}(t) = g(t)\,\delta_{T_s}(t)$$



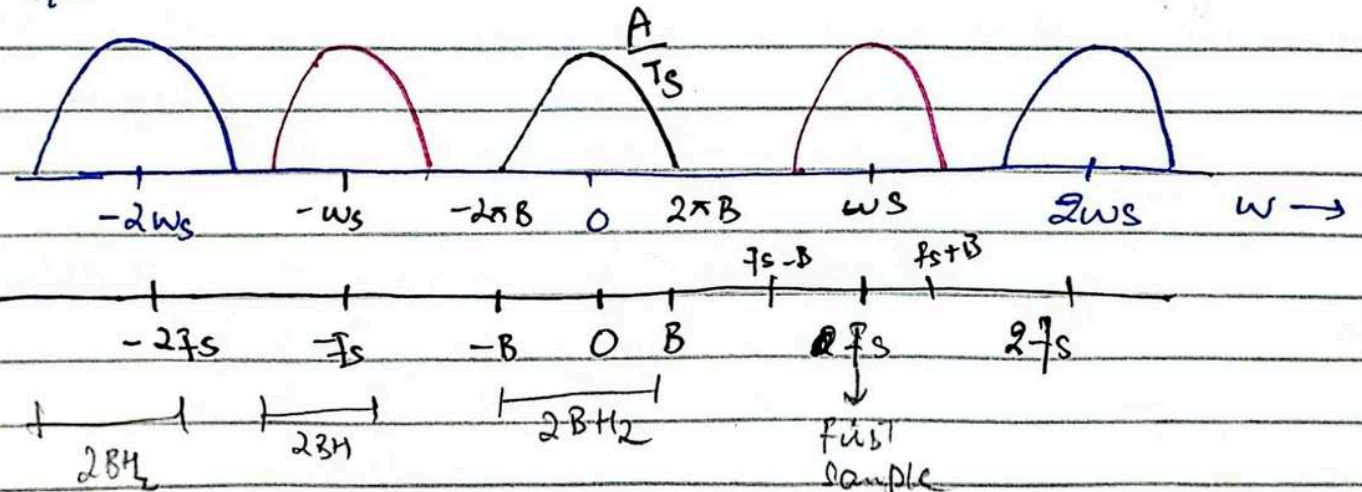
Original signal will not exist only impulse chain will exist.

$$\bar{g}(t) = g(t)\,\delta_{T_s}(t)$$

$$\bar{g}(t) = \frac{1}{T_s}\left[\underbrace{g(t)}_{G(\omega)} + \underbrace{2g(t)\cos\omega_s t}_{G(\omega-\omega_s)+G(\omega+\omega_s)} + \underbrace{2g(t)\cos 2\omega_s t}_{G(\omega-2\omega_s)+G(\omega+2\omega_s)} + \cdots \;\; \cdots\cdots\right]$$

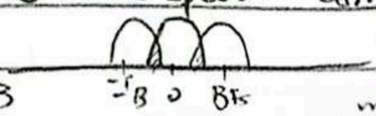
$$\bar{G}(\omega) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} G(\omega - n\omega_s)$$

$\bar{G}(\omega)$



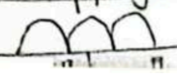
$R > 2B$

if it become less Than 2B these spectrum come closer Than 2B the spectrum start overlapping



$f_s < 2B$

$f = 2B$



When we get overlap this mean spectrum is lost $G(\omega)$ does not exist it lost its significant part

$\bar{g}(t)$ is samples

To reconstruct $g(t)$ from $\bar{g}(t)$, $G(\omega)$ needs to be recovered from $\bar{G}(\omega)$. This is only possible when there is no overlap between successive cycles of $\bar{G}(\omega)$ so

$$f_s > 2B \qquad \left[\text{or } T_s < \frac{1}{2B}\right]$$

Nyquist Rate $f_s = 2B$

Nyquist Interval $T_s = \dfrac{1}{2B}$

The minimum sampling frequency $f_s = 2B$ required to recover $g(t)$ from its sample $\bar{g}(t)$ is called the Nyquist Rate for $g(t)$. and the corresponding sampling interval $T_s = \frac{1}{2B}$ is called the nyquist interval

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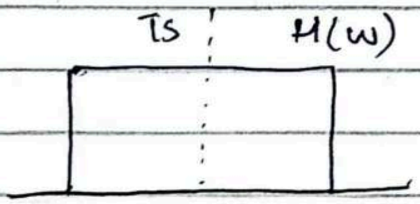
Lecture 23

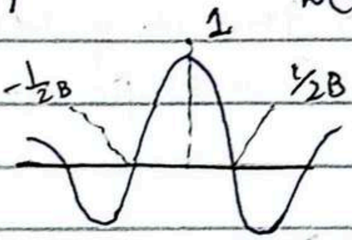
Signal Reconstruction (mean we only have sample of signal no real signal we need real signal)

Signal reconstruction is possible by passing the sampled signal through an ideal LPF of Bandwidth B Hz and gain T_s. The filter's transfer function is

$$\underset{\text{filter}}{H(\omega)} = T_s \, rect\left(\frac{\omega}{4\pi B}\right)$$



IFT $\Longleftrightarrow$



→ magnitude of this is same as sample

we would get sample points when sample is pass through filter

$\therefore \; h(t) = 2BT_s \, (2\pi Bt)$

If sampled at exactly the nyquist rate i.e. $2BT_s = 1$ than

$$h(t) = sinc(2\pi Bt)$$

When the sampled signal $\bar{g}(t)$ is input to $H(\omega)$ the output is $g(t)$

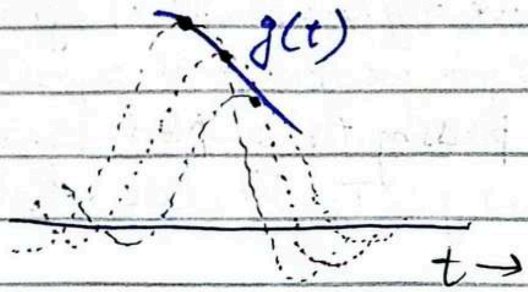
$$g(t) = \begin{cases} \sum_k g(kT_s)\, h(t - kT_s) \\ \sum_k g(kT_s)\, sinc(2\pi B(t - kT_s)) \\ \sum_k g(kT_s)\, sinc(2\pi Bt - k\pi) \end{cases}$$

→ observe $h(t) = 0$ at all Nyquist sampling instants $\left(t = \pm \frac{n}{2B}\right)$

except at $t = 0$

Each sample in $\bar{g}(t)$ being an impulse generates a sinc pulse of height equal to the strength of the sample

at every zero crossing there is only 1 pulse



Interpolation formula yields value of $g(t)$ between samples as a weighted sum of all sample values

Practical Difficulties:

Ideal LPF is non realisable non causal so sampling at exactly nyquist rate $f_s = 2B$ is not good enough and sampling frequency $f_s > 2B$ needs to be used.

nyquist rate
↳ min sampling rate 2B

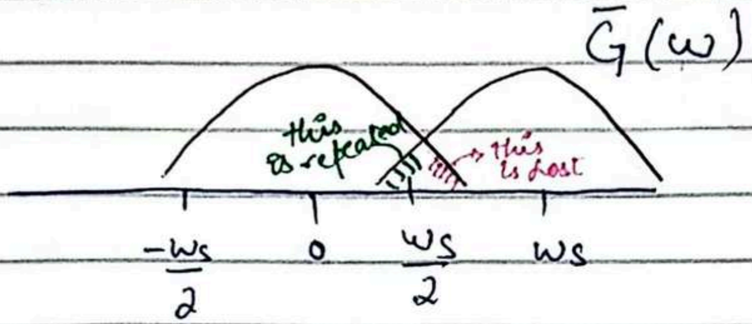
Aliasing (also a pratical difficulty)

Sampling theoram proved with an assumption that the signal $g(t)$ is band limited

→ All pratical real Life signal are time limited and thus can never be band limited

→ The $\overline{G(w)}$ consist of overlapping cycle of $G(w)$ repeating every f_s Hz
↳ sampling frequency

Asymtotically decaying
not bandlimited



Aliasing effect : 1, Loss of tail of $G(\omega)$ beyond $|f| > f_s/2$

2, Reappearance of this inverted back to overlap

Anti Aliasing Filter

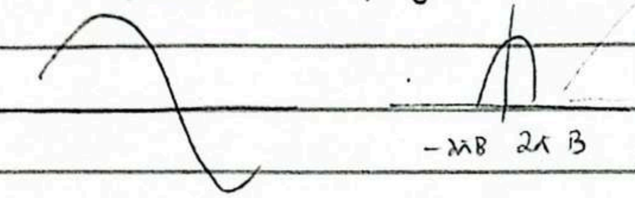
Supression of higher frequencies [beyond $f_s/2$] can be accomplished by a LPF of bandwidth $f_s/2$ Hz

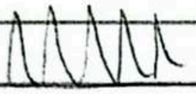
→ Anti aliasing filtering is performed prior to sampling

18-11-2024

Lecture 24

Practical sampling





real pulse train

↳ now successive cycle is not of same magnitude — effect of using practical sampling
↳ successive cycle in decaying

Maximum Information rate: we can transmit two pieces of error free information of per Hertz bandwidth

Applications

* PAM (pulse amplitude modulation) Amplitude vary
* PWM (pulse width modulation) width vary (of pulse)
  ↳ +ve value (wider pulse) → sample value Large more width
  → at zero crossing neither wider nor thinner (normal)
  ↳ -ve value (thinner pulse)
* PPM (pulse position modulation) pulse changes its position w.r.t sample value
* TDM (time division multiplexing)
* PCM (pulse code modulation) three process need to be done
  - Sample (on time axis)
  - Quantization (on amplitude axis)
  - Code assigning

Advantages of digital Communication

↳ More rugged than analoge communication
↳ viability of regenerative repeatre (major advantage)
↳ You transmitted certain sample some noise added of 50km of distance at every 50km repeater (knows original amplitude remove noise added gives original back) (helpful to transmit source Large distance)

Analoge (noise cannot be removed)

(3rd Last book Dim Pg 280)

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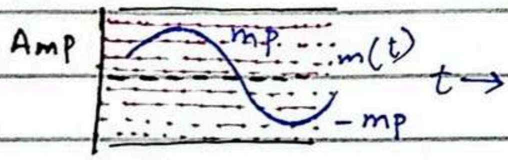
Lecture:

Quantization An analog signal can be converted into a digital signal by means of sampling and quantization.

Amplitude of the analog signal $m(t)$ lie in the range $(-m_p, m_p)$ which is



Quantization take place on amplitude axis
Sampling take place on time axis

partioned into 'L' subintervals each of magnitude

$$\Delta V = \frac{2m_p}{L}$$

Each of the sample is approximated to one of the 'L' numbers

The signal is digitized with quantized samples taking an argue of the "L" values Such a signal is an "L array digital Signal"

Quantization Error The quantization error Lies on the range $\left(-\frac{\Delta V}{2}, \frac{\Delta V}{2}\right)$ where

$$\Delta V = \frac{2m_p}{L}$$

Assuming the error is equally Likely to Lie anywhere in the range $\left[-\frac{\Delta V}{2}, \frac{\Delta V}{2}\right]$ the mean square quantizing error $\tilde{q}^2$ is given by

$$\tilde{q}^2 = \frac{1}{\Delta V}\int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} q^2 \, dq$$

$$= \frac{1}{\Delta V}\left| \frac{q^3}{3} \right|_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}}$$

$$\tilde{q}^2 = \frac{\Delta V^2}{12}$$

$$\tilde{q} = \frac{mp^2}{3L^2}$$

$\tilde{q}^2(t)$ is the power of quantization noise we denote it by N_q

$$N_q = \tilde{q}^2(t) = \frac{mp^2}{3L^2}$$

here mp is constant

Reconstructed signal

when L is increase bandwidth requirement increase
Quantizer 16 level 4 bit is require to transmit (16)²
32 than 5 bit is used to

$$\hat{m}(t) = m(t) + q(t)$$

$$S_o = \tilde{m}^2(t)$$

$$N_o = N_q = \frac{mp^2}{3L^2}$$

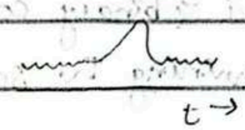
$$\boxed{\frac{S_o}{N_o} = \frac{3L^2\,\tilde{m}^2(t)}{m_p^2}}$$

mp is the peak amplitude value that a quantizer accept and is therefore a constant of quantizer

The output signal to noise ratio is a linear function of the message

Signal power

Speed signal characteristics smaller amplitude value zada hoti higher kaam hoti



max value has lower amplitude

Non Uniform Quantization

$$SNR \propto \widetilde{m^2(t)}$$

error introduce is equal chahay wo higher sa ho ya lower se

talker to talker variations are as much as 40dB

Statistically, the smaller amplitudes pre-dominate in speed SNR is often low (increase)

$$\text{Quantization error} = \frac{(\Delta V)^2}{12} \propto \text{Step size}$$

Quantizer is fixed L is const cannot change m(t) is also const

When we want to increase SNR ratio amplitude ↑ ΔV↑ quantization error ↑ here signal to noise ratio is dec

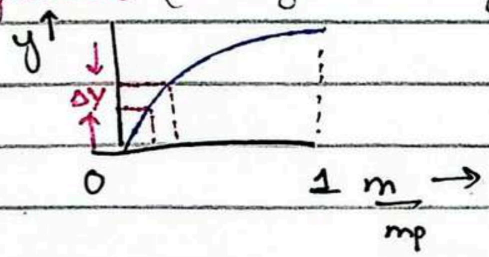
amplitude ↓ ΔV↓ quantization error ↑ SNR ratio↑

Either — Non Uniform Quantizer

or — Compressor then uniform quantization

μ Law A-Law

Compression (message ka dose y become smaller)



$\mu = 100$ for $L = 128$

$\mu = 255$ for $L = 256$

In non uniform L would not change it remain const

Two compressions law are considered μ Law and A Law

μ Law

$$y = \frac{1}{\ln(1+\mu)} \ln\left(1 + \frac{\mu m}{m_p}\right) \qquad 0 < \frac{m}{m_p} < 1$$

A Law

$$y = \begin{cases} \dfrac{A}{1+\ln A}\left(\dfrac{m}{m_p}\right) & 0 < \dfrac{m}{m_p} \le \dfrac{1}{A} \\ \dfrac{1}{1+\ln A}\left(1 + \ln A\left(\dfrac{Am}{m_p}\right)\right) & \dfrac{1}{A} \le \dfrac{m}{m_p} \le 1 \end{cases}$$

A television signal video & audio has a bandwidth of 4.5MHz This signal is sampled quantized & binary coded to obtain a PCM signal

a) determining the sampling rate if the signal is sampled at 20% above the nyquist rate

b) If the samples are quantized in 1024 Levels determine the binary pulses require to encode each sample

c) Determine the binary pulse rate of the binary coded signal and the minimum bandwidth require to transmit this signal

a) Nyquist Rate $= 2B = 2(4.5 \times 10^6) = 9MHz$

Sampling rate 20% above the Nyquist rate $=$ 120% of $9MHz = 10.8\ MHz$

$= 10.8M$ samples/sec

b) $L = 1024$

$2^{10} = 1024$

We require 10 binary pulses for each level

c) Sampling Rate $= 10.8 \times 10^6$

10 binary pulses per sample

Binary pulse rate $= 10.8 \times 10^6 \times 10$

$= 108$ Mbit/s

Minimum Bandwidth $= \frac{108}{2} = 54MHz$

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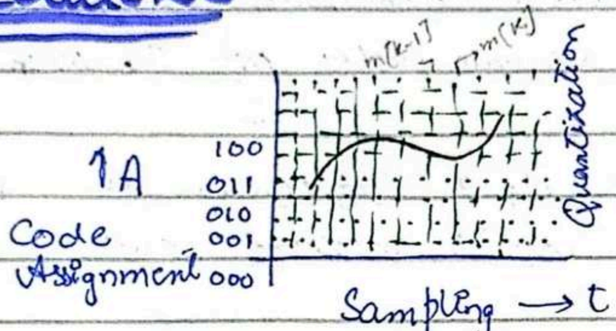
Lecture 26

for 1 sample need 3 bit



$L = 8$

$2^n = L$

$\boxed{n = 3}$

To do Pulse code modulation we require following 3 steps

1. Sampling (on time axis)
2. Quantization (on amplitude axis)
3. Code Assignment (transmitted by signal on output)

→ easiest and simple
→ most famous digital modulation scheme

Differential Pulse Code Modulation (DPCM):

→ transmit difference b/w the sample

Analog messages have an inherent redundancy. Proper exploitation of this redundancy leads to encoding a signal with lesser number of bits. $m[K]$ is the K^{th} sample

Instead of transmitting $m[k]$, Lets transmit the difference

$$d[k] = m[K] - m[k-1]$$

d[k] have smaller value than m[k]

At the receiver knowing the $d[k]$ and the previous sample $m[k-1]$ we can reconstruct $m[k]$

By transmitting the difference $d[k]$ the peak amplitude m_p of transmitted values is reduce considerably

m_p — now is actually d_p → peak of the differences

Since the quantization interval $\Delta V = \frac{m_p}{L}$ → peak of the message

reduced by a factor $\frac{d_p}{m_p}$

For a given L the reduction in mp reduces the quantization interval ΔV thus the quantization noise is reduced by a square factor $\frac{\Delta V^2}{12}$ The ΔV in DPCM

is reduced by factor of $\frac{dp}{mp}$ where dp is differential peak and mp is message peak

Advantage: able to reduce error by a significant amount

DPCM can be further improved if we may estimate $\tilde{m}[k]$ based on previous sample and transmit

→ estimate

$$d[k] = m[k] - \tilde{m}[k]$$

using Taylor Series → first derivative → second derivative

$$m(t+T_s) = m(t) + T_s \dot{m}(t) + \frac{T_s^2}{2!} \ddot{m}(t) + \cdots$$

This is an approximation based on first two values

$$m(t+T_s) \approx m(t) + T_s \dot{m}(t)$$

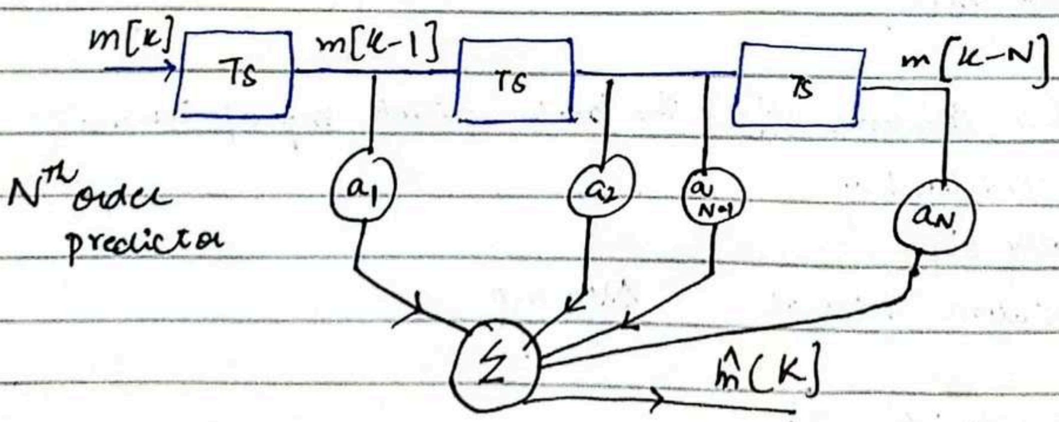
$$m[k+1] \simeq m[k] + T_s \left[\frac{m[k] - m[k-1]}{T_s} \right]$$ → Sampling interval

$$m[k+1] \simeq 2m[k] - m[k-1]$$

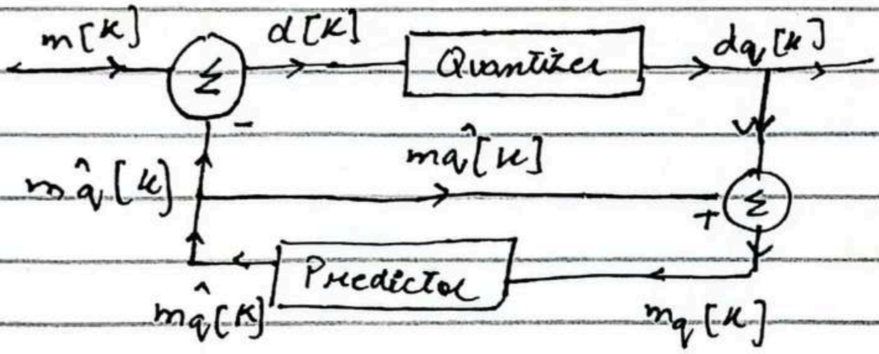
So the generic prediction formula

$$\hat{m}[k] \simeq a_1 m[k-1] + a_2 m[k-2] + \cdots a_N m[k-N]$$

Delay Line Transversal Filter (Implementation of approximation)



In real DPCM, $m[K]$ can not be reconstructed only $m_q[K]$ can be since $m_q[K-1]$, $m_q[K-2]$, ... $d_q[K]$ will be available



Modulator (DPCM)

$$d[K] = m[K] - \hat{m_q}[K]$$

$$d_q[K] = d[K] + q[K] \text{ error} \rightarrow [A]$$

$$m_q[K] = \hat{m_q}[K] + d_q[K]$$

$$= m[K] - d[K] + d_q[K]$$

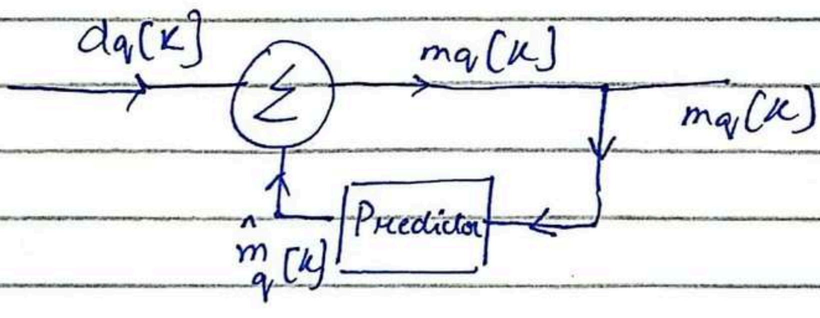
$$= m[K] - d[K] + d[K] + q[K]$$

$$m_q[K] = m[K] + \underbrace{q[K]}_{\text{error}}$$

↳ Quantize version of $m[K]$

the noise & error associated with $d[K]$ is very less than $m[K]$)

DPCM Demodulator



Analog & Digital Communication

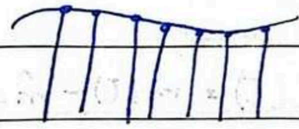
Date: 27 NOV

Lecture:

End Problem
Ch 6
6.1-4, 6.1-6, 6.2-3,
6.2-4, 6.2-5, 6.2-8
6.2-9
7 Q's + Sampling related Problem

Delta Modulation (DM)

Sample correlation used in DPCM is further exploited in delta modulation (DM) by oversampling (typically 4 times the Nyquist rate) the baseband signal



DM is typically a 1 bit DPCM that is a DPCM that uses only two levels ($L=2$) for the generation of $m[k] - \hat{m_q}[k]$

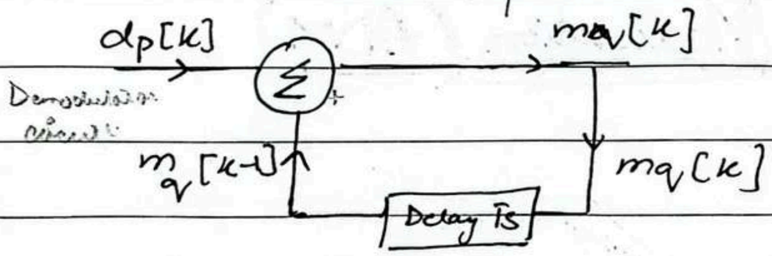
We do oversampling which increase correlation between the sample

$L = 2$

$2^n = L$

$\boxed{2^1 = 2}$

DM uses first order predictor



Only 0 and 1 can be transmit
If previous sample is larger than 0 is transmitted
If previous sample is smaller than 1 is transmitted

$$m_q[k] = m_q[k-1] + d_q[k] \rightarrow (A)$$

$$m_q[k-1] = m_q[k-2] + d_q[k-1] \rightarrow (B)$$

Replace from (B) to (A)

$$m_q[k] = m_q[k-2] + d_q[k-1] + d_q[k] \rightarrow (C)$$

$$\vdots$$

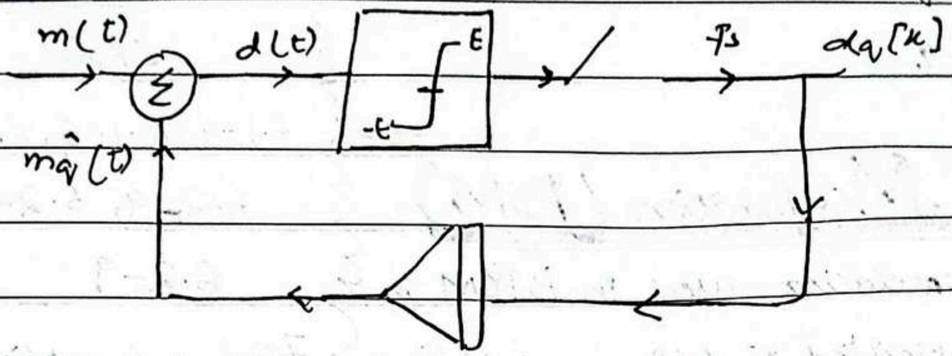
$$m_q[k] = \sum_{m=0}^{k} d_q[m] \rightarrow (D)$$

Delta Modulation / 1 bit transmit

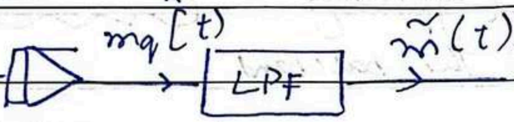
Date: ____________

DM Modulator

comparator (Threshold sa upar E transmit, aus sa neeche to -E transmit)



DM Demodulator



$d(t) = m(t) - \hat{m_q}(t)$ is the "signal error" which is applied to a comparator

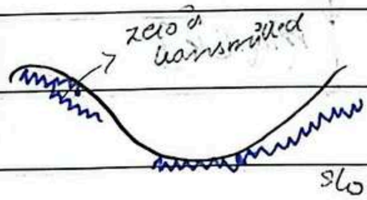
If $d(t)$ is positive, comparator o/p is $+E$

If $d(t)$ is negative, comparator o/p is $-E$

→ related to step size

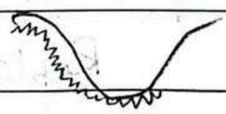
Adaptive Delta Modulation (ADM)

Step size is adopt
jasay signal overload
ma jatay ga step
size increase kar
dy ga



error occur.

slope over load
if slope easily followed
but if slope is like



if continous 1 we understand that it is continously increasing Then we increase step size bcz error is coming if step size the error is reduced

if continous 0 we understand that it is continously decreasing than we increase step size bcz error is coming and we reduce error by increasing step size

Date:

variation in m(t) (message signal) smaller than the step value (threshold of coding) are lost in delta modulation

Slope overload causes dq [K] to have several pulses of same polarity in succession. (If signal is decreasing than continously zero is tranmitting and if signal is increasing continously one is transmitting

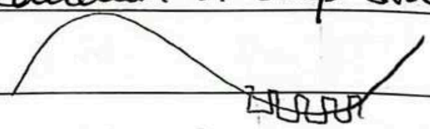
This requires change in step size The adaptive delta modulation (ADM) varies the step size to counter the slope overloading effect and follow the signal m(t) more closely

signal is either decending more speedly continously zero is tranmitting so to reduce error we increase step size Signal is either ascending more speedly continously one is tranmitting so to reduce error step size is increase

01 01 01

The pulses in dq [K] alternating continously in polarity indicates small amplitude variation and requires a reduction in step size

In ADM above Two pattern of pulses are



reconize and the step size is automatically adjusted

if 0000 ya 1111 is transmitted step size increase

if 010101 is transmitted Than step size decrease